

Arithmetic ground-state geometry

Physical caps, arithmetic sources and finite-dimensional residuals

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Abstract

We investigate positive supersymmetric theories and explicit auxiliary responses as sources of the finite-interval Weil form. Interacting prime models retain Euler dependence, while a genuine shape deformation produces a three-state sector inside a nine-vacuum theory. Compact physical caps and opposite-twist reflection fix the residue normalization and a homogeneous middle-class Euler metric. We derive both metric endpoints, including the contributions of escaping vacua, and prove uniqueness conditional on the stated physical chiral equation. Separately, an explicit gamma preparation is closed on the full logarithmic source domain. Robin gluing, coherent delays and positive prime-return graphs expose the contact terms and residuals that their positive realizations require. A joint response, using an independently proved Neumann high-mode bound, replaces an infinite-rank compact error by an explicit positive finite-rank error while retaining all mixed terms. The complete arithmetic sign at each fixed support length is thereby equivalent to a finite matrix inequality. Its positivity and compatibility under unbounded support enlargement remain open; the positive completion is not a proof of Weil positivity or of the Riemann hypothesis.

1 Introduction: the project and what has been learned

The purpose of this project is to explain an arithmetic positivity statement by constructing a system whose positivity is already understood. The arithmetic object is the Weil quadratic form associated with the Riemann zeta function. Its nonnegativity on every smooth compactly supported complex input is equivalent to the Riemann hypothesis. The desired argument would identify this form with an ordinary positive bulk energy or Hilbert-space norm. Numerical tests could then check the identity; they would not be the source of its sign.

This manuscript develops one route toward such a construction. It studies supersymmetric ground states, their parameter-dependent metrics, and interacting theories that retain prescribed gamma and prime factors. It produces exact local-factor identities and exclusions of specified repairs. It does not produce the full Weil norm identity. Understanding both the successes and their limits is essential: a positive quantum Hamiltonian can encode an arithmetic factor without making the required arithmetic quadratic form positive.

1.1 The arithmetic quantity that the theory must reproduce

Let f be a smooth complex function supported in $I_L = (-L/2, L/2)$, and let F be its zero extension to $\mathbb{R}$. Use the Fourier convention

$$\widehat{F}(\tau) = \int_{\mathbb{R}} F(x) e^{-i\tau x} dx.$$

In the normalization used here, the target has the decomposition

$$\begin{aligned} Q_L[f] = & K[F] + w_0 \|f\|^2 + 2|C(f)|^2 - 2|S(f)|^2 \\ & - 2 \sum_{\substack{p \text{ prime}, m \geq 1 \\ m \log p < L}} (\log p) p^{-m/2} \operatorname{Re} \langle F, U_{m \log p} F \rangle. \end{aligned} \quad (1.1)$$

Here $U_d F(x) = F(x - d)$ compares the profile with a translated copy, and

$$C(f) = \int_{I_L} f(x) \cosh(x/2) dx, \quad S(f) = \int_{I_L} f(x) \sinh(x/2) dx.$$

The two squared global amplitudes constitute the signed pole contribution. The gamma kinetic energy and its local normalization are

$$K[F] = \frac{1}{2\pi} \int_{\mathbb{R}} B(\tau^2) |\widehat{F}(\tau)|^2 d\tau, \quad w_0 = \psi(1/4) - \log \pi < 0, \quad (1.2)$$

$$B(s) = \operatorname{Re} \psi(1/4 + i\sqrt{s}/2) - \psi(1/4), \quad s \geq 0,$$

where $\psi = \Gamma'/\Gamma$ is the digamma function. The distances $m \log p$ and weights $(\log p)p^{-m/2}$ are fixed arithmetic data. A finite support length activates only finitely many such distances.

The gamma kinetic term is nonnegative, but the contact constant, translated correlations and pole contribution cannot be discarded. The project requires the complete sum in (1.1) for all inputs and arbitrarily large L , rather than a chosen finite collection of functions or a fixed list of primes at every length. The Weil criterion and modern operator formulations provide the mathematical setting [1]. Section 2 supplies the form domain and the identities used below; no separate background document is needed to read this manuscript.

1.2 What a positive bulk or quantum realization would mean

In a classical auxiliary-field construction, one fixes the source f , minimizes a nonnegative bulk energy over responding fields u , and seeks

$$Q_L[f] = \inf_u \mathcal{E}_L[f, u], \quad \mathcal{E}_L[f, u] \geq 0. \quad (1.3)$$

The bulk may consist of fields on an extra coordinate, a collection of channels, or other auxiliary degrees of freedom. Its defining action and source map must be specified independently of the unknown positivity of Q_L .

The operator version seeks a positive physical Hilbert space and a linear preparation map $\mathcal{A}_L$ with

$$Q_L[f] = \|\mathcal{A}_L f\|_{\mathcal{H}}^2. \quad (1.4)$$

Equivalently, the associated form operator would satisfy $W_L = \mathcal{A}_L^* \mathcal{A}_L$ in the appropriate form sense. This replaces separate tests of individual functions by one operator identity. The requirement that the identity hold on the full input domain remains. Defining $\mathcal{A}_L$ through an assumed positive square root of W_L would simply assume the desired result.

Supersymmetry offers a way to organize such a physical construction. A nilpotent charge Q with its actual Hilbert adjoint gives

$$H = QQ^\dagger + Q^\dagger Q \geq 0.$$

Its zero-energy states are harmonic representatives: states annihilated by both charges. Their ordinary norms can be nonzero even though their energies vanish. Thus the proposed observable in a supersymmetric ground-state construction is a prepared norm or boundary pairing, not merely the ground-state energy. A positive Hamiltonian alone does not establish (1.4). Quantizing an auxiliary system is useful only if the quantum state space, interactions or gluing law supplies an identity absent from the classical minimization.

1.3 The starting success and the remaining discrepancy

The gamma kinetic term already has an exact positive auxiliary-field realization. The identity

$$B(s) = \sum_{k \geq 0} \frac{2}{a_k} \frac{s}{s + a_k^2}, \quad a_k = 2k + \frac{1}{2},$$

expresses it as a positive tower of resolvent responses. Each response comes from minimizing a squared mismatch and a positive gradient energy. This is a substantial first success: an entire continuous operator, on its logarithmic form domain, is represented by a positive field energy. It does not include the contact or poles.

A second starting model is a positive supersymmetric Morse system tensored with discrete prime sectors. For a finite prime set $\mathcal{S}$ it has a state Ψ_σ and a self-adjoint observable X satisfying

$$\langle \Psi_\sigma, e^{i\tau X} \Psi_\sigma \rangle = \Lambda_{\mathcal{S}}(\sigma + i\tau), \quad \Lambda_{\mathcal{S}}(z) = \pi^{-z/2} \Gamma(z/2) \prod_{p \in \mathcal{S}} (1 - p^{-z})^{-1}. \quad (1.5)$$

The formula is verified by an elementary integral and geometric sums. Its normalized logarithmic derivative yields a positive gamma-plus-prime jump energy $J_{\mathcal{S},L}$. The complete target, however, is

$$W_L = J_{\mathcal{S},L} - \kappa_{\mathcal{S}} I + P_L, \quad \kappa_{\mathcal{S}} > 0, \quad (1.6)$$

when $\mathcal{S}$ includes all primes active on the interval. The missing negative contact and signed pole operator are therefore precise, rather than unspecified corrections. A positive characteristic function or jump process does not by itself account for them.

1.4 Why enlarge the theory and admit interactions?

The original arithmetic state family is too restrictive for the desired extended supersymmetry. Its chosen complex extension has a nonflat scalar metric, whereas an ordinary one-vacuum chiral tt^* system has zero curvature. Here tt^* denotes the compatibility equations relating a physical vacuum metric, its projected connection, and operators induced by variations of a holomorphic superpotential. More vacua permit matrix commutators and hence additional geometry. The underlying fields and dynamics must actually support those equations; changing the name of a two-charge model does not create four physical supercharges.

We therefore allow new bosons, fermions, boundary data and genuine interactions. The models studied below are supersymmetric quantum mechanics with complex coordinates and a holomorphic superpotential, often called Landau–Ginzburg models. Their fermions are represented by differential forms, and their Hamiltonian positivity follows from physical adjoints. A polynomial superpotential can create several isolated vacua and a nontrivial matrix of ground-state overlaps. A periodic complex coordinate introduces an additional global issue: some parameter insertions become multivalued, so ordinary chiral tt^* equations cannot automatically be applied without the periodic or flavor data [6].

Gaussianity is consequently a modeling choice, not the boundary between possible and impossible arithmetic constructions. Quadratic models are valuable controls because they can be solved exactly. An interacting theory is more useful only when its interaction changes the pairing or constraints relevant to (1.4). The calculations here give a concrete example of that distinction.

1.5 Overview of the results

The first part constructs exact local-factor controls. A complex cylinder supports four physical supercharges and retains a gamma period. Quadratic prime fields give the Euler period and a

physical metric proportional to $|1 - q|^{-2}$. A cross-coupled quartic prime theory has nine vacua. At fixed coupling its admissible identity periods cannot be Euler factors; scaling the interaction with the square of the mass restores

$$\mathcal{P}_c(M) = C_c/M, \quad h_c(M, \bar{M}) = H_c/|M|^2, \quad M = 1 - q, \quad C_c, H_c > 0. \quad (1.7)$$

This mass variation is a field dilation and vanishes in the Jacobi ring. Finite-rank curvature and scalar completion tests explain why these identities do not determine the missing arithmetic contact and poles.

The shape deformation supplies a nonzero chiral operator and a three-dimensional symmetry sector. Its residue algebra and period equation are explicit. Exact radially rules out a constant physical-to-period ratio on any complex-open shape region. Compact Koszul–Dolbeault representatives and long-cylinder projection then construct genuine physical caps. Opposite-twist reflection fixes the pairing to $4\pi^2$ times the residue in our Euclidean convention. A normalized homogeneous middle-class cap has exactly the Euler metric. The raw identity period is a different functional and does not descend to the Jacobi ring.

Both metric endpoints can be controlled. At the homogeneous endpoint, non-Morse Hodge comparison and form continuity leave one positive scale. At the massive endpoint, local Gaussian upper bounds and opposite-twist lower bounds keep all nine vacua and fix the full limiting matrix. A subharmonic trace argument proves uniqueness with these endpoints, conditional on the exact physical chiral equation in the stated frame.

The second part addresses the full arithmetic source space. The gamma residual tower is a closed source on the logarithmic form domain, with a maximal interval adjoint, a smooth compact-support core and two controlled singular regulators. The complete form is a bounded signed perturbation of its kinetic operator. A Neumann–Dirichlet comparison fixes the logarithmic eigenvalue scale and proves that an explicit finite-codimension sector is positive.

The pole and prime constructions retain their full costs. Attractive Robin gluing has an unstable auxiliary mode and a reduced odd negative mode when $L > 4$. Remote coherent delays produce the exact pole kernel with a compulsory contact. A nonnegative quantum graph produces every repetition of a prime with a further contact, whose stationary lower bound is sharp. Finite-delay rigidity excludes a specified class of attempts to remove those costs without changing the continuous kernel.

Finally, feedback through the assembled positive source first gives the complete Weil form plus a positive compact error. A common response through all gamma and prime channels, iterated on the proved positive high sector, gives the stronger identity

$$\langle \Phi_L f, \Phi_L g \rangle = Q_L(f, g) + \langle P_N f, D_N P_N g \rangle, \quad D_N > 0. \quad (1.8)$$

Here P_N is an explicitly selected Neumann cosine projection; every low/high coupling is retained. The source is closed on the full logarithmic domain and the error has rank N . The remaining criterion is exactly $S_N = G - D_N \geq 0$ for a specified finite matrix. A rational non-arithmetic control shows why the completion alone does not imply this sign. At $L = 1$ a rational high-mode estimate permits $N = 4$; it does not certify that four-by-four arithmetic matrix.

1.6 Organization and status

Sections 2–7 give the arithmetic target, local quantum theories and shape restrictions. Sections 8–9 construct the physical pairings and endpoint metric data. Sections 10–13 develop the closed source, pole and prime responses and their obstructions. Section 14 gives the finite-rank completion and exact remaining criterion. Section 15 states what is still needed for a full physical positivity law.

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2 The arithmetic target and its positive starting models

Throughout, inner products are conjugate-linear in their first entry. Let $E_L : L^2(I_L) \rightarrow L^2(\mathbb{R})$ denote zero extension. The central Weil form is (1.1); no positivity assumption is made about its sum. For fixed L , all translated and pole terms are bounded forms. Since $B(\tau^2)$ grows logarithmically, its closed semibounded realization has form domain

$$\mathcal{D}_L = \left\{ f \in L^2(I_L) : \int_{\mathbb{R}} \log(2 + |\tau|) |\widehat{E_L f}(\tau)|^2 d\tau < \infty \right\}. \quad (2.1)$$

The positive form K is closed by the Fourier multiplier construction, and bounded perturbations give the closed semibounded target operator W_L . Smooth compactly supported inputs are the initial arithmetic test domain. This operator formulation does not require selected test functions to be estimated individually, but an eventual factorization must hold on the full form domain.

2.1 Gamma kinetic energy as an exact positive response

The elementary partial-fraction expansion of the digamma difference gives

$$B(s) = \sum_{k \geq 0} \frac{2}{a_k} \frac{s}{s + a_k^2}, \quad a_k = 2k + \frac{1}{2}. \quad (2.2)$$

For example, the real part of $\psi(b + it) - \psi(b) = \sum_{k \geq 0} [(k + b)^{-1} - (k + b + it)^{-1}]$ is $\sum_{k \geq 0} t^2 / [(k + b)((k + b)^2 + t^2)]$; taking $b = 1/4$ and $t = \sqrt{s}/2$ proves (2.2).

For a mass $a > 0$, define

$$\mathcal{E}_a[F, u] = \frac{2}{a} \left(\|F - u\|_{L^2(\mathbb{R})}^2 + a^{-2} \|u'\|_{L^2(\mathbb{R})}^2 \right), \quad u \in H^1(\mathbb{R}).$$

Its unique minimizer solves $(1 - a^{-2} \partial_x^2)u = F$. In Fourier space,

$$\widehat{u}(\tau) = \frac{a^2}{a^2 + \tau^2} \widehat{F}(\tau), \quad \inf_u \mathcal{E}_a[F, u] = \frac{1}{2\pi} \int \frac{2}{a} \frac{\tau^2}{a^2 + \tau^2} |\widehat{F}(\tau)|^2 d\tau.$$

Summing these nonnegative channel energies and using monotone convergence gives

$$K[F] = \inf_{(u_k)} \sum_{k \geq 0} \mathcal{E}_{a_k}[F, u_k]. \quad (2.3)$$

The admissible fields have finite displayed energy. On (2.1), the channel minimizers achieve the sum; outside it the infimum is infinite. One must not split off the divergent unprepared sum $\sum_k (2/a_k) \|F\|^2$ before minimizing. Formula (2.3) is the exact positive bulk response used as the starting success.

The same kinetic form has a jump representation. Put

$$n_\gamma(r) = \frac{e^{-|r|/2}}{1 - e^{-2|r|}}, \quad r \neq 0.$$

Expanding this density as $\sum_{k \geq 0} e^{-a_k |r|}$ and using $\int_{\mathbb{R}} e^{-a|r|} (1 - \cos \tau r) dr = (2/a) \tau^2 / (a^2 + \tau^2)$ yields

$$K[F] = \frac{1}{2} \int_{\mathbb{R}} \int_{\mathbb{R}} |F(x) - F(y)|^2 n_\gamma(x - y) dx dy. \quad (2.4)$$

The diagonal singularity is interpreted through the difference square.

2.2 An exact finite-prime ground-state amplitude

Let $\mathcal{S}$ be finite and $\sigma > 0$. On

$$\mathcal{H}_{\mathcal{S}} = L^2(\mathbb{R}, dy) \otimes \bigotimes_{p \in \mathcal{S}} \ell^2(\mathbb{N}_0),$$

use the closed Morse operator and bounded prime differences

$$A_{\sigma/2} = \partial_y + \frac{e^y - \sigma/2}{2}, \quad D_p = T - p^{-\sigma/2} I, \quad (T\xi)_n = \xi_{n+1}.$$

Each defines a two-term complex with its physical Hilbert adjoint; their graded tensor product has a positive supersymmetric Hamiltonian. The Morse kernel is obtained by integrating a first-order equation, while $D_p \xi = 0$ gives a geometric sequence. The adjoint kernels vanish: the Morse adjoint solution grows at $+\infty$, and the zeroth component of $D_p^* \xi = 0$ forces every component to vanish. Consequently the product has a unique bosonic vacuum ray,

$$\Psi_{\sigma}(y, \mathbf{n}) = \pi^{-\sigma/4} \exp\left(\frac{\sigma y/2 - e^y}{2}\right) \prod_{p \in \mathcal{S}} p^{-\sigma n_p/2}. \quad (2.5)$$

Morse supersymmetry and zeta-related Morse Fourier amplitudes have existing antecedents [2, 3]; no priority is claimed for these elementary factor constructions.

The multiplication operator

$$X = \frac{y}{2} - \sum_{p \in \mathcal{S}} n_p \log p - \frac{\log \pi}{2}$$

is self-adjoint on its maximal multiplication domain. Substituting $t = e^y$ and summing geometric series proves

$$\|\Psi_{\sigma}\|^2 = \Lambda_{\mathcal{S}}(\sigma), \quad \langle \Psi_{\sigma}, e^{i\tau X} \Psi_{\sigma} \rangle = \Lambda_{\mathcal{S}}(\sigma + i\tau). \quad (2.6)$$

The state normalization in (2.5) is specified data. The Hamiltonian only determines the ray.

2.3 A positive derivative and the missing contact

The product $\Lambda_{\mathcal{S}}$ is nonzero on $\operatorname{Re} z > 0$. Its logarithmic derivative is

$$\frac{\Lambda'_{\mathcal{S}}(z)}{\Lambda_{\mathcal{S}}(z)} = -\frac{\log \pi}{2} + \frac{1}{2} \psi(z/2) - \sum_{p \in \mathcal{S}} \frac{\log p}{p^z - 1}. \quad (2.7)$$

Taking the real part and subtracting its value on the real axis gives

$$\begin{aligned} \partial_{\sigma} \log \frac{|\Lambda_{\mathcal{S}}(\sigma + i\tau)|^2}{\Lambda_{\mathcal{S}}(\sigma)^2} \Big|_{\sigma=1/2} &= B(\tau^2) \\ &+ 2 \sum_{p \in \mathcal{S}, m \geq 1} (\log p) p^{-m/2} [1 - \cos(m \log p \tau)]. \end{aligned} \quad (2.8)$$

Every term on the right is nonnegative. Define $J_{\mathcal{S},L}$ by compressing this Fourier multiplier through zero extension. If $a_{p,m} = (\log p) p^{-m/2}$, then

$$J_{\mathcal{S},L}[f] = K[E_L f] + \sum_{p \in \mathcal{S}, m \geq 1} a_{p,m} \|E_L f - U_{m \log p} E_L f\|^2.$$

The repetitions are summable for finite $\mathcal{S}$.

Proposition 2.1 (Exact contact decomposition). *Suppose $\mathcal{S}$ contains every prime p with $\log p < L$. Put*

$$\kappa_{\mathcal{S}} = -w_0 + 2 \sum_{p \in \mathcal{S}, m \geq 1} a_{p,m} > 0, \quad P_L[f] = 2|C(f)|^2 - 2|S(f)|^2.$$

On $\mathcal{D}_L$,

$$Q_L[f] = J_{\mathcal{S},L}[f] - \kappa_{\mathcal{S}} \|f\|^2 + P_L[f]. \quad (2.9)$$

Proof. Expand each translated difference square as $2\|F\|^2 - 2\operatorname{Re}\langle F, U_d F \rangle$. If $d \geq L$, the correlation vanishes by disjoint support. The surviving correlations are exactly the prime terms in (1.1); the sum of all the diagonal contributions is cancelled by the definition of $\kappa_{\mathcal{S}}$. $\square$

Additional inactive primes change J and κ by cancelling diagonal quantities. Their inclusion is harmless only when that cancellation is kept. An independent all-prime jump sum is divergent on every nonzero compactly supported input: for sufficiently large p , its first translation has disjoint support and contributes $2(\log p)p^{-1/2}\|f\|^2$. The divergence follows already from the divergence of $\sum_p 1/p$. Thus the finite-prime control is not a justification for an unrestricted positive jump limit.

The normalized squared amplitude in (2.8) is a positive-definite function before differentiation, but its logarithmic derivative is a different observable. Equation (2.9), rather than the positivity of the characteristic function, is what specifies the missing arithmetic problem.

2.4 The vacuum metric is not the squared amplitude

A chosen complex extension of (2.5) is

$$\Psi_z = e^{zX/2} e^{-e^y/2}, \quad h(z, \bar{z}) = \|\Psi_z\|^2 = \Lambda_{\mathcal{S}}(\operatorname{Re} z).$$

With $\sigma = \operatorname{Re} z$, the second derivative of $\log \Lambda_{\mathcal{S}}(\sigma)$ is the variance of X in the normalized state, hence strictly positive. In a holomorphic line frame,

$$\partial_{\bar{z}} \partial_z \log h = \frac{1}{4} (\log \Lambda_{\mathcal{S}})''(\sigma) > 0.$$

In contrast, $|\Lambda_{\mathcal{S}}(z)|^2$ has zero logarithmic curvature wherever $\Lambda_{\mathcal{S}}$ is holomorphic and nonzero. Confusing these two metrics erases precisely the geometric obstruction motivating the enlargement.

3 A positive four-supercharge enlargement

An enlargement of the field space provides a concrete way to obtain four real supercharges without retaining the spectrum of a two-supercharge model. The construction below adds an angular coordinate to the real Morse variable. It retains the gamma function as a holomorphic period, but its physical vacuum is a differential form on the enlarged target. Adding two complex oscillators then produces a physical Euler-factor metric in an explicitly specified parameter frame. These statements concern positive quantum theories and identifiable observables; they do not identify their vacuum pairing with the complete Weil form.

3.1 The cylinder and its supersymmetry algebra

Let

$$\mathcal{X} = \mathbb{R} \times (\mathbb{R}/2\pi\mathbb{Z}), \quad Y = y + i\theta, \quad \mathcal{H} = L^2\Omega^*(\mathcal{X}; \mathbb{C}), \quad (3.1)$$

with flat metric $dy^2 + d\theta^2$ and the ordinary positive inner product on complex differential forms. Write E_y, E_θ for exterior multiplication by $dy, d\theta$, and I_y, I_θ for their Hilbert adjoints. Thus

$\{E_i, I_j\} = \delta_{ij}$, while the anticommutators of two exterior multiplications or two contractions vanish. Initially take $a > 0$ and define

$$\mathcal{W}_a(Y) = \frac{1}{2}(e^Y - aY), \quad h = \text{Re } \mathcal{W}_a = \frac{1}{2}(e^y \cos \theta - ay). \quad (3.2)$$

Although $\mathcal{W}_a$ has an additive period, h is single valued for real a . In particular $\Delta h = h_{yy} + h_{\theta\theta} = 0$.

On smooth compactly supported periodic forms define

$$\begin{aligned} Q_1 &= E_y(\partial_y + h_y) + E_\theta(\partial_\theta + h_\theta), \\ Q_1^\dagger &= I_y(-\partial_y + h_y) + I_\theta(-\partial_\theta + h_\theta), \\ Q_2 &= -E_y(\partial_\theta - h_\theta) + E_\theta(\partial_y - h_y), \\ Q_2^\dagger &= I_y(\partial_\theta + h_\theta) + I_\theta(-\partial_y - h_y). \end{aligned} \quad (3.3)$$

Equivalently, $Q_1 = d + dh \wedge$ and $Q_2 = d^c - d^c h \wedge$, where $d^c = E_\theta \partial_y - E_y \partial_\theta$. All adjoints in (3.3) refer to (3.1); an indefinite fermionic pairing is not involved.

For a general smooth real h , the exterior relations give

$$\begin{aligned} Q_1^2 &= Q_2^2 = 0, \quad \{Q_1, Q_2^\dagger\} = 0, \\ \{Q_1, Q_2\} &= -2(\Delta h)E_y E_\theta, \\ \{Q_1, Q_1^\dagger\} - \{Q_2, Q_2^\dagger\} &= 2(\Delta h)(N - 1), \quad N = E_y I_y + E_\theta I_\theta. \end{aligned} \quad (3.4)$$

For example, the coefficient of $E_y E_\theta$ in the mixed nilpotent anticommutator is $[\partial_y + h_y, \partial_y - h_y] + [\partial_\theta + h_\theta, \partial_\theta - h_\theta] = -2\Delta h$. The remaining identities follow by the same exterior expansion. Harmonicity therefore gives a common Hamiltonian $H = \{Q_1, Q_1^\dagger\} = \{Q_2, Q_2^\dagger\}$ and four real charges

$$\begin{aligned} R_1 &= Q_1 + Q_1^\dagger, & R_2 &= i(Q_1 - Q_1^\dagger), & \{R_i, R_j\} &= 2\delta_{ij}H. \\ R_3 &= Q_2 + Q_2^\dagger, & R_4 &= i(Q_2 - Q_2^\dagger), \end{aligned} \quad (3.5)$$

Their common square is

$$\begin{aligned} H &= -\partial_y^2 - \partial_\theta^2 + \frac{1}{4}|e^Y - a|^2 \\ &\quad + e^y [\cos \theta (E_y I_y - E_\theta I_\theta) - \sin \theta (E_y I_\theta + E_\theta I_y)]. \end{aligned} \quad (3.6)$$

This is an exponential bosonic interaction with a Hessian coupling to the fermions. In the corresponding normal-ordered coherent-state description its Euclidean action is

$$S_E = \int dt \left\{ \frac{1}{4}(\dot{y}^2 + \dot{\theta}^2) + |\nabla h|^2 + \bar{c}_i \dot{c}_i + 2\bar{c}_i h_{ij} c_j \right\}. \quad (3.7)$$

The normalization follows from the kinetic operator $-\Delta$ in (3.6); the scalar normal-ordering contribution $-\Delta h$ vanishes. Positivity is the operator statement $H \geq 0$, not positivity of every fermionic path-integral integrand.

3.2 Closed operators and the physical vacuum

The displayed algebra has a closed realization. Each R_i is a Dirac-type first-order operator with smooth Hermitian multiplication potential on a complete flat manifold. Choose compact-support cutoffs χ_L tending to one with $|d\chi_L| \leq C/L$. The potential commutes with the cutoff,

so $\|[R_i, \chi_L]\| \leq C/L$. A deficiency vector satisfying $R_i^* \psi = \pm i\psi$ is locally regular by ellipticity. Integration by parts with $\chi_L^2 \psi$, followed by taking the imaginary part, gives

$$\|\chi_L \psi\|^2 \leq \frac{C}{L} \|\chi_L \psi\| \|\psi\|. \quad (3.8)$$

Consequently both deficiency spaces vanish. This proves essential self-adjointness on the stated common core. The same argument applies to every nonzero real linear combination of the R_i .

On that core the graph norms of the four charges coincide, by (3.5). Their closures therefore have a common quadratic-form domain, and their squares define the same nonnegative self-adjoint H . Polarizing the identities for real linear combinations proves the mixed algebra as a form identity. Since each closed R_i commutes with the spectral resolution of $R_i^2 = H$, all products in (3.5) are defined on $\text{Dom } H$, where the operator identities follow. This argument also applies to the product targets used below. It is compatible with the noncompact Witten-operator framework of [7]; the cutoff argument itself does not require the vacuum-counting theorem.

Restricting the function h to $\theta = 0$ gives $(e^y - ay)/2$, but restricting a function is not a quantum reduction. For example, the scalar real-line operator obtained from $A = \partial_y + (e^y - a)/2$ is

$$A^\dagger A = -\partial_y^2 + \frac{1}{4}(e^y - a)^2 - \frac{1}{2}e^y. \quad (3.9)$$

The scalar cylinder Hamiltonian has no negative Hessian term because $\Delta h = 0$. In degree one its Hessian term has eigenvalues $\pm e^y$, rather than $-e^y/2$. A delta function at a fixed angle is not a state of (3.1). Moreover, e^{-h} grows exponentially near $\theta = \pi$ as $y \rightarrow +\infty$, and e^h grows near $\theta = 0$. Neither supplies a normalizable scalar or top-form vacuum.

Proposition 3.1. *For every $a \in \mathbb{C} \setminus \{0\}$, the cylinder Hamiltonian with periodic differential forms has exactly one complex normalizable zero-energy state. It is a one-form. The assertion uses the well-tame Morse inequalities of [7] at large Witten deformation, together with the continuation argument below; it does not compute a vacuum norm in an arithmetic source frame.*

Proof. First let $a > 0$ and replace h by Th , $T > 0$. There is exactly one critical point, $(y, \theta) = (\log a, 0)$, of Morse index one. The complete flat cylinder has bounded geometry, and

$$\liminf_{|y| \rightarrow \infty} |dh| = \frac{a}{2} > 0, \quad \frac{|\text{Hess } h|}{|dh|^2} \rightarrow 0. \quad (3.10)$$

At the negative end the Hessian tends to zero and $|dh| \rightarrow a/2$; at the positive end $|dh|$ grows like $e^y/2$ while the Hessian grows like e^y . These are the well-tame hypotheses in [7]. Their strong Morse inequalities, including equality of the full alternating sums, give graded Dirac index -1 for sufficiently large T . This step uses the analytic theorem, rather than an assumed identification of critical points with exact quantum states. It does not require the additional Thom–Smale comparison theorem.

The graded Dirac operator remains Fredholm for all $T > 0$. At $y \rightarrow -\infty$ its square tends to $-\partial_y^2 - \partial_\theta^2 + T^2 a^2/4$; at the other end its potential tends to $+\infty$, since $T^2 |dh|^2$ dominates the Hessian. These estimates are uniform on compact positive T intervals and give a positive lower bound outside a compact set. Local elliptic compactness then gives Fredholmness. The estimate $|\text{Hess } h| \leq \varepsilon |dh|^2 + C_\varepsilon$ also makes the graph domains equivalent on those intervals, so the family is continuous in graph norm and its index is constant. This is the required continuation from the large-deformation Morse calculation [2, 7].

In degrees zero and two the squared operator is simply $-\Delta + T^2 |dh|^2$. A zero state would have both vanishing weak gradient and vanishing product with $|dh|$, and hence must be zero. Thus the index -1 implies precisely one degree-one vacuum.

For complex a , the globally defined one-form dh is described explicitly below. Its variation with a is a bounded constant one-form. The negative-end gap becomes $T^2|a|^2/4$ and the positive-end estimate persists. Any path in $\mathbb{C} \setminus \{0\}$ therefore gives a Fredholm continuation from positive real a . The even-degree kernels remain zero because $\Delta h = 0$ locally. The same count follows at $T = 1$. At $a = 0$ the end gap closes, so this argument makes no assertion there. $\square$

Writing the vacuum as $\omega = P dy + Q d\theta$, its equations are

$$\begin{aligned} \partial_y Q - \partial_\theta P + h_y Q - h_\theta P &= 0, \\ -\partial_y P - \partial_\theta Q + h_y P + h_\theta Q &= 0. \end{aligned} \tag{3.11}$$

These determine the physical state on the full cylinder. No scalar gamma wavefunction is substituted for their solution.

3.3 Periodic parameters and the gamma period

For $a = a_R + ia_I$ one has locally

$$h = \frac{1}{2}(e^y \cos \theta - a_R y + a_I \theta), \quad h(y, \theta + 2\pi) - h(y, \theta) = \pi a_I. \tag{3.12}$$

Although h need not be globally a function, its differential is a smooth closed real one-form on the cylinder. The supercharges use this one-form and its complex-structure rotation, so remain globally defined on periodic forms. Their definition is not globally a conjugation by a single-valued e^h . Similarly, $\mathcal{W}_a(Y + 2\pi i) - \mathcal{W}_a(Y) = -\pi i a$, and the proposed chiral insertion $\partial_a \mathcal{W}_a = -Y/2$ is multivalued. The local holomorphic dependence of a superpotential thus does not by itself supply an ordinary globally defined rank-one chiral tt^* problem. Periodic Landau–Ginzburg theories with deck and flavor data provide a relevant framework [6]; any such extra data must be included explicitly in a proposed arithmetic pairing.

The ordinary Berry connection of the specific one-vacuum realization above is locally flat. Indeed, H has real coefficients for every complex a , and its isolated one-dimensional kernel admits a real normalized generator ψ locally. Reality and differentiation of $\langle \psi, \psi \rangle = 1$ imply $\langle \psi, d\psi \rangle = 0$. A possible global sign holonomy does not change this local conclusion. Parameter connections with additional insertions or flavor structure are separate constructions; the distinction matters when using supersymmetric Berry equations [5].

There is nevertheless an exact holomorphic gamma period:

$$\mathcal{P}_\gamma(a) = \int_{Y \in \mathbb{R}} e^{-2\mathcal{W}_a(Y)} dY = \int_0^\infty t^{a-1} e^{-t} dt = \Gamma(a), \quad \operatorname{Re} a > 0. \tag{3.13}$$

The factor two in the exponent is part of the definition. For complex a the real path is a convergent relative cycle, not necessarily a steepest-descent thimble. Such periods belong to the holomorphic boundary quantities of Landau–Ginzburg theory [6]. Equation (3.13) is not an evaluation of the Hermitian norm of the physical one-form in (3.11).

3.4 Two complex oscillators and a physical Euler metric

Introduce independent parameters a, q with $\operatorname{Re} a > 0$, $|q| < 1$, put $M = 1 - q$, and enlarge the target to $\mathcal{X} \times \mathbb{C}^2$ with its product flat metric. The superpotential is

$$\mathcal{W}_{a,q}(Y, U, V) = \frac{1}{2} \left[e^Y - aY + \frac{M}{2}(U^2 + V^2) \right]. \tag{3.14}$$

This introduces two further complex bosons and their form-valued fermions. On a flat Kähler target the preceding construction extends with d^c defined by the complex structure. The required Hessian condition is pluriharmonicity, rather than merely a vanishing trace; it holds for $h = \operatorname{Re} \mathcal{W}_{a,q}$. Alternatively, the graded tensor product of the three two-dimensional complexes verifies the four-charge algebra directly. The common-core closure argument above applies on this complete product target. In particular,

$$|\nabla h|^2 = \frac{1}{4}|e^Y - a|^2 + \frac{|M|^2}{4}(|U|^2 + |V|^2). \quad (3.15)$$

Since $\operatorname{Re} M > 0$, elementary integration on the real cycle gives

$$\frac{1}{2\pi} \int_{\mathbb{R}^3} e^{-2\mathcal{W}_{a,q}} dY dU dV = \frac{\Gamma(a)}{1-q}. \quad (3.16)$$

For a fixed prime p , setting $a = z/2$, $q = p^{-z}$ and specifying the source multiplier $\pi^{-z/2}$ produces $\pi^{-z/2}\Gamma(z/2)(1-p^{-z})^{-1}$ for $\operatorname{Re} z > 0$. Neither this restriction of parameters nor its Fourier variable $\operatorname{Im} z$ identifies an arithmetic coordinate with physical Euclidean time.

The oscillator vacuum metric is separately computable. For one field with $\mathcal{W}_U = MU^2/4$, write $M = re^{i\phi}$ and $U' = e^{i\phi/2}U = x' + iy'$. Then

$$h_U = \frac{r}{4}(x'^2 - y'^2), \quad \psi_M = e^{-r|U|^2/4} dy', \quad \|\psi_M\|^2 = \frac{2\pi}{r}. \quad (3.17)$$

The form ψ_M satisfies $Q_1\psi_M = Q_1^\dagger\psi_M = 0$. Equivalently, the scalar oscillator ground energy is r , cancelled by the occupied negative Hessian direction, whose contribution is $-r$. Every other oscillator state has positive energy, proving uniqueness. Consequently the product theory also has a unique physical vacuum for $aM \neq 0$.

To make the parameter frame explicit, define

$$\begin{aligned} s_M &= (2\pi)^{-1/2} e^{-i\phi/2} \psi_M \\ &= \frac{e^{-r|U|^2/4}}{2i\sqrt{2\pi}} \left(dU - \frac{\bar{M}}{|M|} d\bar{U} \right). \end{aligned} \quad (3.18)$$

The second expression is single valued for $M \neq 0$. If P_M denotes the physical ground-state projector, then $P_M \partial_{\bar{M}} s_M = 0$. To verify this without identifying different Hodge realizations, normalize the real state as $\chi_M = \sqrt{r/(2\pi)} \psi_M$. It has zero local Berry connection, and $s_M = M^{-1/2} \chi_M$ on a local branch. The coefficient is holomorphic, which proves the assertion directly in the Hilbert space of real differential forms complexified. Thus

$$\langle s_M, s_M \rangle = |M|^{-1}, \quad g_{UV} = |M|^{-2} = |1 - q|^{-2}. \quad (3.19)$$

No identification with a twisted-Dolbeault identity class is needed for this explicit calculation.

The coefficient in (3.18) fixes the source normalization. In the associated tensor frame the full physical metric is $g_\gamma(a, \bar{a})|1 - q|^{-2}$, where the cylinder factor has not been evaluated. It is distinct from the period (3.16). Furthermore, $\partial_M \partial_{\bar{M}} \log g_{UV} = 0$ away from zero, and multiplication of the two-field frame by M gives unit metric. Thus the denominator has meaning relative to the specified source; positivity alone does not select it. At $M = 0$ oscillator confinement and its normalizable vacuum disappear. These facts provide a positive physical benchmark, while leaving the arithmetic contact and full pole pairing undetermined.

3.5 An affine gamma–prime coupling fails two tests

A direct attempt to couple the sectors replaces their constant mass by $M + \lambda e^Y$:

$$\mathcal{W}_{a,q}^\lambda = \frac{1}{2} \left[e^Y - aY + \frac{M + \lambda e^Y}{2} (U^2 + V^2) \right]. \quad (3.20)$$

For real $a, M, \lambda > 0$ its real-cycle integral converges, and integrating U, V gives

$$\begin{aligned} \mathcal{P}_\lambda(a, M) &= \int_0^\infty \frac{t^{a-1} e^{-t}}{M + \lambda t} dt < \frac{\Gamma(a)}{M}, \\ \frac{d}{d\lambda} \mathcal{P}_\lambda \Big|_{0+} &= -\frac{\Gamma(a+1)}{M^2}. \end{aligned} \quad (3.21)$$

The inequality and the right derivative are exact analytic checks of the amplitude. They are not numerical positivity bounds.

There is a separate obstruction in the full complex target. At $t_0 = -M/\lambda \neq 0$, the critical equations admit

$$e^Y = t_0, \quad U^2 + V^2 = \frac{2(a - t_0)}{\lambda t_0}. \quad (3.22)$$

This is a noncompact complex critical quadric: writing $(U + iV)(U - iV)$ as the displayed constant exhibits points escaping to infinity. Thus a convergent real contour conceals nonisolated classical vacua elsewhere in the physical field space. The isolated-vacuum hypotheses of a standard massive ground-state construction cannot simply be assumed. This observation does not by itself prove a vanishing quantum spectral gap.

The period obstruction has a useful scoped extension. Fix q and suppose a mass $m(t, q)$ is nonzero on $t > 0$ and satisfies the integrability assumptions for uniqueness of its Mellin transform. If

$$\int_0^\infty t^{a-1} e^{-t} m(t, q)^{-1} dt = \frac{\Gamma(a)}{1 - q} \quad (3.23)$$

holds throughout an open Mellin strip for independently variable a , then $m(t, q) = 1 - q$ almost everywhere on that contour. Indeed, subtract the two integrands, fix a real part $\operatorname{Re} a = b$ in the strip, and put $t = e^x$. The resulting L^1 function of x has identically zero Fourier transform, so vanishes almost everywhere. This excludes a nonconstant quadratic mass preserving the entire factorized period within this class. It does not exclude a match only on the linked locus $q = p^{-2a}$, a different insertion or boundary problem, or a nonquadratic prime theory. Those alternatives require additional structure beyond the positive product Hamiltonian constructed here.

4 Interacting prime fields: an exclusion and an exact Euler metric

The quadratic prime fields give an exact control, but only one vacuum. We now replace them by the cross-coupled polynomial

$$F_{M,g,\lambda}(U, V) = \frac{M}{2}(U^2 + V^2) + \frac{g}{4}(U^4 + V^4) + \frac{\lambda}{2}U^2V^2. \quad (4.1)$$

The prime superpotential is $\mathcal{W} = F/2$; adjoining the cylinder gives $\mathcal{W}_{\text{full}} = \frac{1}{2}[e^Y - aY + F]$. All physical target metrics in this section are flat. The factor $1/2$ matters: the holomorphic period considered below has weight $e^{-2\mathcal{W}} = e^{-F}$, while the physical vacuum metric is defined by the supersymmetric Hilbert problem.

4.1 Nine isolated physical vacua

For definiteness take $M \neq 0$, $g > 0$ and $\lambda = g/2$. The critical equations are

$$U(M + gU^2 + \lambda V^2) = 0, \quad V(M + gV^2 + \lambda U^2) = 0.$$

There is one origin, four axis points with the nonzero coordinate squared equal to $-M/g$, and four points with

$$U^2 = V^2 = -M/(g + \lambda).$$

The Hessian determinant is M^2 at the origin, $-2M^2(1 - \lambda/g)$ at an axis point, and $4(g^2 - \lambda^2)U^2V^2$ at a point with both coordinates nonzero. All nine points are nondegenerate.

The homogeneous cubic gradient of the quartic part has no zero on the complex unit sphere when $g \neq 0$ and $g^2 \neq \lambda^2$. Compactness of that sphere gives constants with

$$|\nabla F| \geq c_1 R^3 - c_2 R, \quad |\text{Hess } F| \leq c_3(1 + R^2).$$

The gradient square therefore dominates every fixed multiple of the Hessian at infinity. This verifies strong tameness on the complete Kähler–Stein target $\mathbb{C}^2$. The corresponding Landau–Ginzburg Hodge theorem identifies the physical vacuum space with the Jacobi classes in middle degree [8]; its dimension here is nine. This use of the theorem supplies existence and the cohomological identification, not a formula for the metric.

At fixed g, λ , the superpotential-variation observable is multiplication by $[\partial_M \mathcal{W}] = [(U^2 + V^2)/4]$. Its critical-point values are 0, $-M/(4g)$ and $-M/[2(g + \lambda)]$, with multiplicities 1, 4, 4. In the twisted-Dolbeault convention specified below, the Higgs operator is half this variation. It is still nonscalar, so this interacting theory has room for nontrivial matrix geometry. Whether its observable retains the arithmetic factor is a separate test.

4.2 Every admissible identity period obeys a differential equation

Let Γ_M be a locally flat family of rapid-decay relative cycles, with no finite-field boundary and sufficient decay for the following polynomial moments and integrations by parts. The coefficients of a transported cycle are fixed in relative homology; arbitrary constant complex combinations are allowed. Parameter-dependent fitted coefficients, singular insertions and nonvanishing boundary terms are not part of this class. The usual thimble description is one realization of these assumptions [10].

Write

$$\langle P \rangle_M = \frac{1}{2\pi} \int_{\Gamma_M} P(U, V) e^{-F} dU \wedge dV, \quad s = U^2 + V^2, \quad t = U^2 V^2,$$

and set $I = \langle 1 \rangle_M$, $A = \langle s \rangle_M$, $B = \langle t \rangle_M$. Differentiation keeps g, λ fixed and gives

$$I' = -A/2, \quad A' = -\langle s^2 \rangle_M/2, \quad B' = -\langle st \rangle_M/2.$$

The exact two-forms obtained by differentiating Ue^{-F} and Ve^{-F} , and then UV^2e^{-F} and VU^2e^{-F} , have vanishing boundary integrals. Their expansions are

$$2I - MA - g\langle U^4 + V^4 \rangle_M - 2\lambda B = 0, \quad A - 2MB - (g + \lambda)\langle st \rangle_M = 0.$$

Consequently

$$\begin{aligned} I' &= -A/2, \\ A' &= -I/g + MA/(2g) - (g - \lambda)B/g, \\ B' &= -A/[2(g + \lambda)] + MB/(g + \lambda). \end{aligned} \tag{4.2}$$

For $g \neq \lambda$ one may eliminate the moments using

$$A = -2I', \quad B = \frac{2gI'' - MI' - I}{g - \lambda}.$$

Substitution yields

$$2g(g + \lambda)I''' - (3g + \lambda)MI'' + [M^2 - (3g + \lambda)]I' + MI = 0.$$

Only the subsystem probed by the identity insertion and mass derivatives has rank at most three; this does not reduce the full nine-dimensional vacuum space to three dimensions.

Theorem 4.1 (Fixed-coupling period exclusion). *Fix $g > 0$ and $\lambda = g/2$. Every identity period in the preceding cycle class obeys*

$$\mathcal{L}_g I := 6g^2 I''' - 7gMI'' + (2M^2 - 7g)I' + 2MI = 0. \quad (4.3)$$

No such period can equal a nonzero constant multiple of $1/M$ on a nonempty open parameter region.

Proof. The equation is the specialization of (4.2). Direct substitution gives

$$\mathcal{L}_g(1/M) = -\frac{g(7M^2 + 36g)}{M^4}, \quad (4.4)$$

which is not identically zero on any open region. Every constant combination of admissible periods obeys the same linear equation, so cancellation among saddle contributions cannot change the conclusion. $\square$

The coefficients of (4.3) are entire in M and its leading coefficient is nonzero. Hence every local solution continues to an entire solution in the finite M -plane. Fixed quartic confinement persists at $M = 0$ even though critical points collide; the identity period cannot acquire the Gaussian Euler pole there. Stokes changes of an asymptotic cycle basis do not create a pole in a transported solution. This is stronger than the strict suppression of the real-cycle integral, which follows immediately from the positive real quartic term.

4.3 A mass-dependent interaction restores the period exactly

The fixed-coupling hypothesis is substantive. Let $c > 0$ be constant and choose

$$g(M) = cM^2, \quad \lambda(M) = cM^2/2. \quad (4.5)$$

With

$$G_c(A, B) = \frac{1}{2}(A^2 + B^2) + \frac{c}{4}(A^4 + B^4 + A^2B^2),$$

one has the holomorphic change-of-variables identity

$$F_M(U, V) = G_c(\sqrt{M}U, \sqrt{M}V). \quad (4.6)$$

For positive real M use the real cycle. For nonzero complex M use its dilation transport on a local square-root branch. Then

$$I_c(M) = \frac{C_c}{M}, \quad C_c = \frac{1}{2\pi} \int_{\mathbb{R}^2} e^{-G_c(A, B)} dA dB > 0. \quad (4.7)$$

The equality follows from $dA \wedge dB = M dU \wedge dV$, with no parameter-dependent fitted cycle coefficients. The fixed-coupling differential equation does not apply, because total differentiation now includes $g'(M)$ and $\lambda'(M)$.

For each $M \neq 0$, the theory remains strongly tame and has nine nondegenerate critical points. At $M = 0$, all coefficients in F_M vanish and confinement is lost. The eight nonzero critical points move to infinity like $M^{-1/2}$. The degeneration gives a physical location for the recovered pole.

4.4 The interacting physical metric has the same mass dependence

We specify the physical frame by the identity holomorphic volume class. The strongly tame Landau–Ginzburg Hodge identification gives its middle-degree harmonic representative, using the fixed normalization $\mathcal{W} = F/2$. The bridge to the twisted Dolbeault realization is explicit in the charge conventions of Section 3:

$$D = \frac{Q_1 - iQ_2}{2} = \bar{\partial} + \frac{1}{2}\partial\mathcal{W}\wedge, \quad \{D, D^\dagger\} = H/2. \quad (4.8)$$

Thus it has the same physical harmonic kernel. Fan’s differential $\bar{\partial} + \partial\mathfrak{f}\wedge$ corresponds here to $\mathfrak{f} = \mathcal{W}/2 = F/4$. Its canonically normalized Higgs operator is $[\partial_M\mathfrak{f}] = \frac{1}{2}[\partial_M\mathcal{W}]$. This constant scaling does not change the Jacobi ideal, but it fixes the normalization of any future nonzero curvature equation. All comparisons below use this same physical realization; a scalar period is not declared to be a norm.

Proposition 4.2 (Interacting physical Euler dependence). *Let α_c be the physical harmonic representative of the identity class $dA \wedge dB$ for the reference theory $G_c/2$, and put $H_c = \|\alpha_c\|^2 > 0$. For the family (4.6), the identity-class metric in the (U, V) volume frame is*

$$h_c(M, \bar{M}) = \frac{H_c}{|M|^2}, \quad M \neq 0. \quad (4.9)$$

Proof. Let $\phi_M(U, V) = (\sqrt{M}U, \sqrt{M}V)$. It pulls back the reference flat metric to $|M|$ times the flat metric and pulls back the superpotential to the one at M . For a constant metric rescaling $g \mapsto ag$, a degree-one twisted exterior differential has adjoint and Laplacian

$$d_h^{\dagger, ag} = a^{-1}d_h^{\dagger, g}, \quad \Delta_{h, ag} = a^{-1}\Delta_{h, g}.$$

The same degree-counting statement holds in the twisted Dolbeault realization of the Jacobi identification. Thus constant rescaling changes no harmonic kernel. Pullback is consequently compatible with the harmonic representative and its holomorphic volume class.

In real dimension four, the norm of a degree-two form satisfies $\|\beta\|_{ag}^2 = a^{4/2-2}\|\beta\|_g^2 = \|\beta\|_g^2$. It follows that ϕ_M^* is an isometry on the middle-degree harmonic spaces when both original theories use the flat metric. Since $\phi_M^*(dA \wedge dB) = M dU \wedge dV$, uniqueness and linearity of harmonic representatives give

$$\alpha_M = M^{-1}\phi_M^*\alpha_c,$$

which proves (4.9). Reversing the square-root branch composes with $(A, B) \mapsto (-A, -B)$. This preserves the potential, metric and identity volume class, so this source is unchanged; other classes may have monodromy. $\square$

For $M = 1 - q$, division by the fixed reference norm gives exactly $|1 - q|^{-2}$. This is a physical interacting Euler metric. It does not identify H_c with $|C_c|^2$. A single source rescaling changes a period linearly and a norm quadratically, and therefore cannot independently set both constants to one. The normalized period and metric ratios at $M = 1$ are exact without that extra assertion.

4.5 Why this interaction does not yet constrain the contact

Differentiate (4.6) along the family (4.5). The chain rule gives

$$\partial_M F_M = \frac{U\partial_U F_M + V\partial_V F_M}{2M}. \quad (4.10)$$

The right side belongs to the Jacobian ideal. Thus the projected chiral mass operator is zero in $\mathbb{C}[U, V]/(\partial_U F_M, \partial_V F_M)$. In an ordinary chiral tt^* description this makes the curvature in the M direction zero. Correspondingly,

$$\partial_M \partial_{\bar{M}} \log h_c = 0 \quad (M \neq 0),$$

and the holomorphic frame change $\alpha_M \mapsto M \alpha_M$ removes the local denominator. The interaction is genuine at fixed c , but varying M only dilates it. The source frame and the singular degeneration carry the Euler dependence.

Interactions can therefore preserve an exact arithmetic local factor. To constrain the missing pairing one needs more than the field dilation in (4.10). Section 7 provides a genuine shape deformation, and Section 8 constructs a normalized physical source in a different Jacobi class.

5 Finite-rank metric constraints and a semisimple control

Additional vacua can compensate the curvature of an arithmetic ground-state line without introducing negative norms. They do not, however, automatically preserve its physical state family or its arithmetic observable. We establish two finite-rank restrictions and then give a positive rank-two solution of the metric equations that illustrates the distinction.

5.1 The metric equation and its determinant

Let $z = \sigma + i\eta$ be an ordinary chiral parameter. In an intrinsic holomorphic frame, write the positive Hermitian vacuum metric as g and set

$$A_z = g^{-1} \partial_z g, \quad \mathcal{K} = \partial_{\bar{z}}(g^{-1} \partial_z g), \quad C^{\dagger g} = g^{-1} C^* g. \quad (5.1)$$

Here C^* denotes conjugate transpose. Our convention for the ordinary chiral tt^* equation is

$$\partial_{\bar{z}} C = 0, \quad \mathcal{K} = [C, C^{\dagger g}]. \quad (5.2)$$

Indeed $[D_z, D_{\bar{z}}] = -\mathcal{K}$ for $D_z = \partial_z + A_z$, $D_{\bar{z}} = \partial_{\bar{z}}$, and (5.2) gives flatness of $D_z + \zeta^{-1} C$ and $D_{\bar{z}} + \zeta C^{\dagger g}$. The vacuum-metric interpretation originates in topological-antitopological fusion [4]; the parameter-multiplet hypothesis is essential to the supersymmetric Berry-connection derivation [5]. The existence of four formal supercharges alone does not identify a specified parameter family with (5.2).

For a finite set of primes S , the coherent-state benchmark has

$$h(\sigma) = \Lambda_S(\sigma), \quad \Lambda_S(z) = \pi^{-z/2} \Gamma(z/2) \prod_{p \in S} (1 - p^{-z})^{-1}, \quad \operatorname{Re} z > 0. \quad (5.3)$$

Writing $u = \log h$ gives

$$V(\sigma) := u''(\sigma) = \frac{1}{4} \psi_1(\sigma/2) + \sum_{p \in S} \frac{(\log p)^2 p^{-\sigma}}{(1 - p^{-\sigma})^2} > 0, \quad \partial_z \partial_{\bar{z}} \log h(\operatorname{Re} z) = \frac{V}{4}. \quad (5.4)$$

This physical norm is distinct from the squared holomorphic amplitude $|\Lambda_S(z)|^2$.

Proposition 5.1 (Determinant constraint). *For a finite-rank positive solution of (5.2), $\partial_z \partial_{\bar{z}} \log \det g = 0$. Consequently $\det g = |d(z)|^2$ locally for a nonvanishing holomorphic d . In particular, the determinant cannot equal $h(\operatorname{Re} z)^a |d(z)|^2$ for a nonzero real a .*

Proof. The finite-dimensional trace gives

$$\partial_z \partial_{\bar{z}} \log \det g = \text{tr } \mathcal{K} = \text{tr}[C, C^{\dagger_g}] = 0.$$

A real harmonic function is locally the real part of a holomorphic function, which proves the factorization. The proposed determinant instead has logarithmic curvature $aV/4 \neq 0$ by (5.4). $\square$

Thus tensoring a rank- N solution with the original arithmetic line creates determinant curvature $NV/4$. Prescribing h as one component is a different operation: $g = \text{diag}(h, h^{-1})$ is positive and has determinant one. Determinant cancellation is a necessary condition, not the full differential compatibility condition.

5.2 Ambient holomorphicity is a stronger restriction

Proposition 5.2 (Finite ambient-holomorphic obstruction). *Let $\mathcal{H}$ be a fixed positive Hilbert space and let $T(z) : \mathbb{C}^N \rightarrow \mathcal{H}$ be an injective Hilbert-norm holomorphic frame, with $N < \infty$. Set $g = T^*T$ and $P = Tg^{-1}T^*$, the orthogonal projector onto its span. If this induced metric satisfies (5.2), then P is constant on each connected parameter domain and $\mathcal{K} = 0$.*

Proof. With $T' = \partial_z T$, holomorphicity gives $\partial_z g = T^*T'$ and $\partial_{\bar{z}} g = T'^*T$. Hence

$$\mathcal{K} = -g^{-1}T'^*Tg^{-1}T^*T' + g^{-1}T'^*T' = g^{-1}T'^*(1 - P)T', \quad (5.5)$$

$$\text{tr } \mathcal{K} = \left\| (1 - P)T'g^{-1/2} \right\|_{\text{HS}}^2. \quad (5.6)$$

The trace must also vanish by Proposition 5.1. Thus $(1 - P)T' = 0$ and $\mathcal{K} = 0$. Direct differentiation yields $\partial_z P = (1 - P)T'g^{-1}T^* = 0$; taking the adjoint also gives $\partial_{\bar{z}} P = 0$. $\square$

This excludes a direct finite-dimensional holomorphic enlargement of the complete Morse–prime coherent family. In its fixed positive Hilbert space, that family is

$$\Psi_z = e^{zX/2} e^{-e^y/2}, \quad X = \frac{y}{2} - \sum_{p \in S} n_p \log p - \frac{\log \pi}{2}, \quad n_p \in \{0, 1, 2, \dots\}. \quad (5.7)$$

Projection to all $n_p = 0$ sends states at distinct positive real parameters σ_j to nonzero scalar multiples of $e^{\sigma_j y/4} e^{-e^y/2}$. These are linearly independent: division by $e^{-e^y/2}$ and successive differentiation at one point gives a Vandermonde system. Arbitrarily large independent sets exist on every open real interval. Consequently no constant finite-dimensional subspace can contain this family, as Proposition 5.2 would require.

The assumption $\partial_{\bar{z}} T = 0$ must not be substituted for the weaker physical condition $P\partial_{\bar{z}} T = 0$. An intrinsic holomorphic vacuum frame may have nonzero antiholomorphic derivatives in the massive complement. Such derivatives are allowed in physical tt^* geometry and invalidate the positive-trace calculation (5.6). Infinite rank likewise requires separate trace and domain arguments.

The corresponding line-bundle identity makes the remaining freedom explicit. For an intrinsic holomorphic section v , let $h_v = \langle v, v \rangle_g$ and $\beta = (1 - P_L)D_z v$, where P_L projects onto its line. In a holomorphic normal frame, differentiating $\log h_v$ gives

$$\partial_z \partial_{\bar{z}} \log h_v = \frac{\langle v, \mathcal{K}v \rangle_g + \|\beta\|_g^2}{h_v} = \frac{\|C^{\dagger_g} v\|_g^2 - \|Cv\|_g^2 + \|\beta\|_g^2}{h_v}. \quad (5.8)$$

Thus positive line curvature can be supplied by chiral mixing and variation inside a larger vacuum bundle. It is not itself a rank-two obstruction.

5.3 A local positive semisimple metric

Rank-two positivity and reality conditions are studied in [9]. The following explicit construction verifies the metric equations directly; it makes no existence assertion about a microscopic quantum theory.

Proposition 5.3 (Real-axis metric control). *There is a neighborhood of $z = 1/2$ carrying a positive rank-two solution of (5.2), with semisimple C_z , whose first diagonal component on the real axis is $\Lambda_S(\sigma)$.*

Proof. For $r = \operatorname{Re} w > 0$, set

$$g_*(w, \bar{w}) = \begin{pmatrix} \coth(2r) & 0 \\ 0 & \tanh(2r) \end{pmatrix}, \quad C_* = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (5.9)$$

The first curvature entry equals the first commutator entry because

$$\frac{1}{4} \frac{d^2}{dr^2} \log \coth(2r) = \frac{4 \cosh(4r)}{\sinh^2(4r)} = \coth^2(2r) - \tanh^2(2r). \quad (5.10)$$

The second entries have opposite sign. The metric is positive with determinant one, while C_* has distinct eigenvalues ± 1 . For the constant bilinear pairing $\eta_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\eta_0 C_*$ is symmetric and $(\eta_0^{-1} g_*)(\eta_0^{-1} \overline{g_*}) = 1$.

At $\sigma = 1/2$, $h > 1$ and $h' < 0$. Indeed $\Gamma(1/4) > \int_0^1 t^{-3/4}(1-t) dt = 16/5$, $\pi^{1/4} < 2$, and each Euler factor exceeds one. Also $\psi(1/4) = -\gamma - \pi/2 - 3 \log 2 < 0$, while the logarithmic derivatives of the Euler factors are negative. A local branch therefore exists for

$$w(z) = \frac{1}{4} \log \frac{\Lambda_S(z) + 1}{\Lambda_S(z) - 1}, \quad g_{\text{new}}(z, \bar{z}) = g_*(w(z), \overline{w(z)}), \quad C_z = w'(z) C_*. \quad (5.11)$$

After shrinking the neighborhood, $\operatorname{Re} w > 0$ and $w' \neq 0$ there. Holomorphic pullback multiplies both sides of (5.2) by $|w'|^2$. On the real axis $2w = \operatorname{arccoth} h$, so $(g_{\text{new}})_{11}(\sigma, \sigma) = \coth(2w(\sigma)) = h(\sigma)$. The eigenvalues of C_z are the distinct values $\pm w'(z)$. $\square$

The complex extension has changed: $(g_{\text{new}})_{11}(z, \bar{z})$ is generally not $h(\operatorname{Re} z)$. Only a real-axis diagonal norm has been retained. The construction does not specify the physical states (5.7), their complete overlap $\Lambda_S((\bar{z} + w)/2)$, a self-adjoint arithmetic observable X , or a Hamiltonian with the proposed metric as its vacuum metric. Semisimplicity and the displayed algebraic reality condition also do not establish the additional ultraviolet, infrared, and boundary data of a massive quantum field theory. The domain is local, with no all-prime or global half-plane continuation asserted.

Remark 5.4 (Keeping the original complex extension). For $g = \operatorname{diag}(h(\sigma), h(\sigma)^{-1})$ and $\kappa = V/4$, the companion choices $C = \begin{pmatrix} 0 & 1 \\ q & 0 \end{pmatrix}$ and $C = \begin{pmatrix} 0 & q \\ 1 & 0 \end{pmatrix}$ require, respectively,

$$|q(z)|^2 = R_+(\sigma) := h^4 - \kappa h^2, \quad |q(z)|^2 = R_-(\sigma) := \kappa h^{-2} + h^{-4}.$$

Holomorphic q forces $RR'' - (R')^2 = 0$ wherever it is nonzero. Stirling's formula gives $u'' = 1/(2\sigma) + O(\sigma^{-2})$ and superexponential growth of h , whence $(\log R_+)'' = 2/\sigma + O(\sigma^{-2})$ and $(\log R_-)'' = -1/\sigma + O(\sigma^{-2})$ at infinity. Real analyticity on $(0, \infty)$ excludes either identity even on an open patch; identically zero q is excluded by the same large- σ behavior. This is a test of the stated frame and companion normalization, not every rank-two Higgs field. The changed extension in (5.11) avoids these hypotheses.

The control is deliberately permissive: any positive real-analytic norm greater than one with nonzero derivative can be inserted into the same local pullback. In particular, local metric compatibility does not select the arithmetic source normalization or its first normal logarithmic derivative. The next physical construction must preserve an explicitly defined complete pairing, rather than only one metric component, and derive its normalization from the theory. None of the metric results above supplies the Weil pole pairing or proves positivity of the complete Weil form.

6 Completion obstructions and vector-parameter geometry

The positive normalized arithmetic jump energy is an exact control, but its conversion into the Weil form requires a negative contact and a signed pole contribution. We first identify an obstruction to a scalar ground-state completion. We then construct a vector-parameter connection compatible with the benchmark Berry curvature and show why that compatibility does not fix the missing contact. Neither argument excludes positive bulk theories with more general source maps or internal degrees of freedom.

6.1 The complete form and its scalar ground transform

Fix $I = (-L/2, L/2)$ and a finite prime set $\mathcal{S}$ containing every prime with $\log p < L$. For zero extensions $F = E_L f$ and $G = E_L g$, set

$$\begin{aligned} n_\gamma(r) &= \frac{e^{-|r|/2}}{1 - e^{-2|r|}}, & a_{p,m} &= (\log p)p^{-m/2}, & d_{p,m} &= m \log p, \\ \nu_{\mathcal{S}}(dr) &= n_\gamma(r) dr + \sum_{p \in \mathcal{S}, m \geq 1} a_{p,m} (\delta_{d_{p,m}} + \delta_{-d_{p,m}})(dr), \end{aligned} \quad (6.1)$$

$$j(f, g) = \frac{1}{2} \int \nu_{\mathcal{S}}(dr) \int_{\mathbb{R}} \overline{F(x+r) - F(x)} [G(x+r) - G(x)] dx. \quad (6.2)$$

The gamma density is of order $1/(2|r|)$ at zero, and the prime measure has finite mass. Thus j is the nonnegative gamma form plus a bounded perturbation, with the same logarithmic Fourier form domain. Write J for its associated operator. The complete target has the exact form

$$W_L = J - \kappa_{\mathcal{S}} I + P_L, \quad \kappa_{\mathcal{S}} = -w_0 + 2 \sum_{p \in \mathcal{S}, m \geq 1} a_{p,m} > 0, \quad w_0 = \psi(1/4) - \log \pi, \quad (6.3)$$

where

$$p(x, y) = 2 \cosh((x - y)/2), \quad P_L = 2|c\rangle\langle c| - 2|s\rangle\langle s|, \quad c(x) = \cosh(x/2), \quad s(x) = \sinh(x/2). \quad (6.4)$$

Although $p(x, y)$ is positive, P_L is an indefinite rank-two operator. Inactive prime jumps contribute only a diagonal term to j , canceled by $\kappa_{\mathcal{S}}$; adding such primes does not change W_L .

Let $\mu_{\mathcal{S}}$ be the internal jump measure on I^2 obtained from (6.1), and let $k_{\text{esc}}(x) = \int_{x+r \notin I} \nu_{\mathcal{S}}(dr)$. Then j is the internal difference form plus $\int_I k_{\text{esc}} \bar{f}g$. Take a real $h \in C^1(\bar{I})$ bounded above and bounded away from zero. All identities below hold first for $f = hu$, $g = hv$ with $u, v \in C_c^\infty(I)$ and with zero exterior values for h in Jh . Expansion and symmetrization give

$$j(hu, hv) = \frac{1}{2} \int_{I^2} h(x)h(y) (\overline{u(x) - u(y)})(v(x) - v(y)) d\mu_{\mathcal{S}} + \int_I h(Jh) \bar{u}v. \quad (6.5)$$

Consequently the potential $-(Jh)/h$ admits a positive incidence-square representation. This is the useful scalar ground-state mechanism: a negative potential can arise from a positive energy after a

change of ground state. For example, $h = 1$ removes the positive exterior escape rate. Multiplying h by a constant, however, does not change $(Jh)/h$. The normalization of a vacuum ray cannot select the prescribed contact. The incidence operator can be closed to define a positive form; identifying that closure with a prescribed operator sum is a further domain question. Our obstructions use only the displayed compactly supported core.

The pole term has the corresponding identity

$$\begin{aligned} \langle f, P_L g \rangle &= \int_I r_P(x) \bar{f} g \\ &\quad - \frac{1}{2} \int_{I^2} p(x, y) (\overline{f(x) - f(y)}) (g(x) - g(y)) dx dy, \\ r_P(x) &= 8 \sinh(L/4) \cosh(x/2). \end{aligned} \tag{6.6}$$

Combining the two expansions yields the full polarized ground identity

$$\begin{aligned} q_L(hu, hv) &= \frac{1}{2} \int_{I^2} h(x) h(y) (\overline{u(x) - u(y)}) (v(x) - v(y)) [d\mu_S - p(x, y) dx dy] \\ &\quad + \int_I h(W_L h) \bar{u} v. \end{aligned} \tag{6.7}$$

Even a positive solution of $W_L h = 0$ would leave a signed edge measure. The poles change off-diagonal couplings and therefore cannot be treated merely as a local killing potential.

6.2 A sign obstruction before the first prime

Proposition 6.1 (Scalar source obstruction). *Let $\rho > 1$ be the root of $\rho^3 - \rho - 1 = 0$ and put $r_* = \log \rho$. For every $L > r_*$ there are nonzero nonnegative smooth inputs with disjoint supports for which $q_L(u, v) > 0$. Therefore, with a pointwise scalar source identification, q_L cannot be a positive-ground-state transform of a real scalar diffusion or jump form with nonnegative edge weights, even after adding an arbitrary local potential.*

Proof. Away from the finitely many active prime delays, the off-diagonal kernel is

$$R(r) = 2 \cosh(r/2) - n_\gamma(r) = e^{-r/2} \frac{e^{3r} - e^r - 1}{e^{2r} - 1}, \quad r > 0. \tag{6.8}$$

The polynomial $x^3 - x - 1$ is strictly increasing for $x > 1$, so R is positive precisely when $r > r_*$. Choose $d \in (r_*, L)$ away from all active prime delays and two sufficiently narrow patches separated by d . Their difference set avoids those delays and stays above r_* . For nonnegative u, v supported in the respective patches, $q_L(u, v) = \int u(x) R(|x - y|) v(y) dx dy > 0$. The contact and all local potentials vanish on these disjoint inputs. In contrast, a nonnegative scalar jump measure has a nonpositive cross term on disjoint nonnegative inputs; a local diffusion contributes zero. Positive multiplication by h preserves both disjointness and sign. $\square$

The restriction concerns a Markov-type scalar boundary energy, not positivity of an arbitrary Hilbert-space quadratic form. It also persists under scalar reversible bulk elimination. At a finite regulator a positive real symmetric diffusion/jump matrix has nonpositive off-diagonal entries. Its invertible interior block has entrywise nonnegative inverse, so the Schur complement again has nonpositive off-diagonal entries. Limits preserving the bilinear form on these test inputs retain the same sign.

Proposition 6.2 (Scalar phase obstruction). *For $L > 2r_*$, a scalar phase change cannot turn all the off-diagonal couplings into real nonpositive Markov couplings. This obstruction already occurs for some $L < \log 2$.*

Proof. Choose three narrow patches with all pairwise separations above r_* and away from prime delays. Their continuous couplings are positive. A phase change would have to assign phase -1 to each of the three edges. The product of phases around a triangle is 1 for a scalar gauge, whereas the required product is $(-1)^3 = -1$. The argument can be formulated almost everywhere on the patches and does not require pointwise regularity of the phase. Finally, evaluating $x^3 - x - 1$ at $\sqrt{2}$ shows $\rho < \sqrt{2}$, hence $2r_* < \log 2$. $\square$

Matrix states, genuine magnetic holonomy, and nonlocal or derivative source maps are outside these exclusions. The triangle identifies why scalar rephasing is insufficient; it does not exclude an energy containing interference squares with positive cross terms.

6.3 The compulsory diagonal and the contact problem

Indeed the pole kernel has a positive interference-square construction:

$$e_P^+(f, g) = \frac{1}{2} \int_{I^2} p(x, y) (\overline{f(x) + f(y)}) (g(x) + g(y)) dx dy = \langle f, P_L g \rangle + \int_I r_P \bar{f} g. \quad (6.9)$$

The equality follows simply by expanding the square. Its consequence is

$$q_L[f] = j[f] + e_P^+[f] - \int_I (\kappa_S + r_P(x)) |f(x)|^2 dx. \quad (6.10)$$

Thus a positive square reproduces the signed pole operator only together with an additional diagonal energy. The completion must derive the subtraction of this companion as well as the original contact. Adding two pole amplitudes does not by itself perform that task.

More precisely, put $D = \kappa_S I - P_L$. This bounded operator has infinite rank: it equals $\kappa_S I$ on $\{c, s\}^\perp$. A finite-rank correction of the prepared state map cannot remove the entire difference $J - W_L$. This does not exclude a finite number of bulk field species, each carrying infinitely many modes.

The danger of a circular Schur construction is already explicit in a valid one-prime support window. If $\log 2 < L \leq \log 3$, then $D > 0$: $c \perp s$, $\|c\|^2 = L/2 + \sinh(L/2)$, and $\|s\|^2 = -L/2 + \sinh(L/2)$, so its smallest possible exceptional eigenvalue is $\kappa_S - 2\|c\|^2 > 0$. Here $2\|c\|^2 \leq \log 3 + 2/\sqrt{3} < 3$, whereas $\kappa_S \geq -w_0 > 3$. The elementary square completion

$$\mathcal{A}[f, z] = j[f] - 2 \operatorname{Re} \langle D^{1/2} f, z \rangle + \|z\|^2 = q_L[f] + \|z - D^{1/2} f\|^2 \quad (6.11)$$

does have the desired minimized value. But $\mathcal{A} \geq 0$ holds if and only if $q_L \geq 0$. Its positivity is exactly the unknown assertion. Adding $\langle f, Df \rangle$ makes the action manifestly positive and changes its minimum to $j[f]$. A useful coherent constraint must therefore supply an independently derived infinite-dimensional contraction or projection identity, including the pole sectors, rather than repackaging the target's unknown positivity.

6.4 A vector-parameter lift and its normalization freedom

Vector-multiplet parameters offer a different compatibility equation from ordinary chiral tt^* geometry. Four-supersymmetry invariance constrains their Berry connection and projected moment-map operator

by a Bogomolny equation [5]. This is a necessary effective equation, not a converse construction of a microscopic theory.

For the finite-prime benchmark define, on $\text{Re } z > 0$,

$$\Lambda_S(z) = \pi^{-z/2} \Gamma(z/2) \prod_{p \in S} (1 - p^{-z})^{-1}, \quad \ell(z) = \log \Lambda_S(z). \quad (6.12)$$

The logarithm exists and is real on the positive axis. The specified ground family has $\|\Psi_\sigma\|^2 = e^{\ell(\sigma)}$ and normalized extension $\Omega_{\sigma,\eta} = e^{i\eta X/2} \Omega_\sigma$. In the convention $A = i \langle \Omega, d\Omega \rangle$,

$$A_\sigma = 0, \quad A_\eta = -\frac{1}{2} \ell'(\sigma), \quad F_{\sigma\eta} = -\frac{1}{2} \ell''(\sigma), \quad \ell''(\sigma) = \text{Var}_\sigma(X) > 0. \quad (6.13)$$

Introduce an auxiliary third coordinate v , orient parameter space as (σ, η, v) , and define

$$A_\sigma = A_v = 0, \quad A_\eta = -\frac{1}{2} \text{Re } \ell'(\sigma + iv), \quad \Phi = -\frac{1}{2} \text{Im } \ell'(\sigma + iv). \quad (6.14)$$

The Cauchy–Riemann equations give $\nabla \times A = \nabla \Phi$: its σ and v components are, respectively, $-\text{Im } \ell''/2$ and $-\text{Re } \ell''/2$ on both sides. At $v = 0$ this restricts to (6.13), with $\Phi = 0$. This explicit half-space solution works for any suitable holomorphic ℓ real on the positive axis. It neither selects arithmetic factors nor makes the auxiliary extension the actual Berry data of a quantum model.

For completeness,

$$\ell'(z) = -\frac{\log \pi}{2} + \frac{1}{2} \psi(z/2) - \sum_{p \in S} \frac{\log p}{p^z - 1}. \quad (6.15)$$

Its continued local-factor poles have logarithmic-derivative residue -1 , with multiplicities at coincidences. A simple boundary pole produces a two-dimensional dipole-type harmonic Φ , invariant along η , rather than a regular three-dimensional point monopole. Global defects and boundary conditions would require separate physical specifications.

The contact ambiguity remains exact. Replacing $\ell(z)$ by $\ell(z) + cz$, $c \in \mathbb{R}$, corresponds to rescaling the original source Ψ_ζ by $e^{c\zeta/2}$, where $\zeta = \sigma + i\eta$ labels that chosen complex family. Its normalized state changes only by $e^{ic\eta/2}$. Consequently A_η shifts by $-c/2$, a flat pure gauge on the noncompact η -line, while the curvature and Φ stay fixed. Nevertheless

$$\partial_\sigma \log |\Lambda_S(\sigma + i\tau)|^2 \mapsto \partial_\sigma \log |\Lambda_S(\sigma + i\tau)|^2 + 2c. \quad (6.16)$$

Thus curvature cannot determine this scalar contact. Changing the origin of X changes its mean but not its variance; a fixed physical source must supply the missing convention.

At $\sigma = 1/2$ the pole-free arithmetic multiplier is $2 \text{Re } \ell'(1/2 + i\tau) = -4A_\eta(1/2, \tau)$. This equality identifies functions, not a positive norm. The choice $v = \tau$ is an additional proposed correspondence, and the signed pole form is still absent. Scalar completion factors do not restore that form by a pointwise critical-line derivative. A successful enlargement must derive a physical source and the complete pairing, including its contact and residue or boundary contributions. These requirements leave matrix, fermionic, magnetic, and relative-boundary mechanisms open while specifying the precise failure of the scalar and curvature-only repairs tested here.

7 The interaction shape and its three-state metric sector

The mass dilation of Section 4 leaves the chiral mass class zero. A genuine additional parameter is the dimensionless shape $c \in \mathbb{C}^*$ in

$$G_c = \frac{A^2 + B^2}{2} + \frac{c}{4}(A^4 + B^4 + A^2 B^2), \quad f_c = G_c/4, \quad D_c = \bar{\partial} + df_c \wedge. \quad (7.1)$$

The physical family is defined on all of $\mathbb{C}^*$; the real-plane period below has the smaller directly convergent domain $\text{Re } c > 0$.

7.1 Jacobi multiplication, symmetry and residue

Set $s = A^2 + B^2$, $t = A^2 B^2$, and $\Omega = dA \wedge dB$. The Jacobi ideal is generated by $A(1 + cA^2 + cB^2/2)$ and $B(1 + cB^2 + cA^2/2)$. Its nine simple points are the origin, the four axis points with nonzero coordinate squared $-1/c$, and the four points with $A^2 = B^2 = -2/(3c)$. A basis is

$$1, A, B, A^2, AB, B^2, A^2 B, AB^2, A^2 B^2.$$

The invariant subalgebra has the exact relations

$$s^2 = -s/c + t, \quad st = -4t/(3c), \quad t^2 = 4t/(9c^2). \quad (7.2)$$

They follow by reducing the cubic gradient relations; evaluation at the nine points proves independence of the displayed basis.

The group $(\mathbb{Z}_2)^2 \rtimes S_2$ acts by sign changes and exchange. The volume Ω transforms by its determinant character. The corresponding physical symmetry sector therefore consists of invariant polynomials times Ω , and has basis $(1, s, t)\Omega$. The Hamiltonian, harmonic projection and shape insertion preserve this sector. It is a three-dimensional block of the nine-vacuum theory; there is no division of physical norms by the group order eight.

The canonical Higgs insertion is multiplication by

$$[\partial_c f_c] = [(A^4 + B^4 + A^2 B^2)/16] = -s/(16c). \quad (7.3)$$

It has eigenvalues $0, 1/(16c^2), 1/(12c^2)$, and the identity is cyclic under it, since its first two powers generate s and s^2 . Thus a scalar closure of this physical identity sector is insufficient.

Use the holomorphic source frame

$$E = (1, r, v)\Omega, \quad r = cs, \quad v = c^2 t. \quad (7.4)$$

In this frame the multiplication matrix of r and the residue matrix are

$$R = \begin{pmatrix} 0 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & -4/3 \end{pmatrix}, \quad \eta_E = \frac{64}{3} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & -4/3 \\ 1 & -4/3 & 4/9 \end{pmatrix}. \quad (7.5)$$

Here the residue convention is

$$\eta(p\Omega, q\Omega) = \frac{1}{(2\pi i)^2} \int \frac{pq dA \wedge dB}{(f_c)_A (f_c)_B} = \sum_{df_c(a)=0} \frac{p(a)q(a)}{\det \text{Hess } f_c(a)}.$$

The inverse Hessian determinants per point are $16, -16, 12$ on the three types, so the orbit weights are $(16, -64, 48)$. This proves (7.5), including $\eta(1, 1) = 0$ and $\eta_E R = R^T \eta_E$. The orbit idempotents are

$$e_0 = 1 + r + \frac{3}{4}v, \quad e_a = -r - 3v, \quad e_b = \frac{9}{4}v. \quad (7.6)$$

Their residue is $\text{diag}(16, -64, 48)$. A zero identity self-residue does not imply a zero Hermitian norm.

7.2 Radiality and the conditional chiral equation

Let $g_E(c, \bar{c})$ be the physical metric of the fixed sources (7.4). Dilation gives

$$f_c = \phi_c^*((1/c)f_1), \quad \phi_c(A, B) = (\sqrt{c}A, \sqrt{c}B).$$

All three sources become c^{-1} times pullbacks of fixed polynomial volume sources. Middle-degree pullback preserves the L^2 norm under a constant conformal dilation in real dimension four. For the reference differential $\bar{\partial} + u df_1 \wedge$, multiplication of a (p, q) -form by $e^{ip\theta}$ conjugates u to $e^{i\theta}u$. All three volume sources have the same holomorphic degree. Their fixed-source Gram matrix is therefore invariant under this phase. Consequently the entire physical matrix in (7.4) is radial, not only its identity component.

Put $u = 1/c$, $\rho = |u|$, and write this metric as $g(\rho)$. Under the *additional identification* of the physical Hodge family with the exact chiral equation in this source frame, its equations are

$$\partial_{\bar{u}}(g^{-1}\partial_u g) = [C, g^{-1}C^\dagger g], \quad C = R/16, \quad (7.7)$$

$$(g^{-1}g')' + \rho^{-1}g^{-1}g' = \frac{1}{64}[R, g^{-1}R^T g]. \quad (7.8)$$

The factor $1/16$ follows from $f_c = G_c/4$; the displayed physical superpotential variation would be twice this insertion. The Hodge and chiral frameworks are those of [8, 4, 5]. We retain the exact identification in (7.7) as an explicit hypothesis for the metric selection theorem below; Hodge existence alone is not its proof.

The eigenvalues of C are $0, -1/16, -1/12$, the three critical-value scales of uf_1 . Neither these scales nor the algebra fix Stokes data. For example, in the orbit frame a constant positive metric proportional to $\text{diag}(16, 64, 48)$ makes C normal and has zero curvature. With a consistent overall residue normalization it satisfies the reality relation as well. The physical endpoint in Section 9 excludes this constant control in the specified frame.

7.3 The exact shape-period equation

The separate holomorphic period is

$$\mathcal{C}(c) = \frac{1}{2\pi} \int_{\mathbb{R}^2} e^{-G_c(A,B)} dA dB, \quad \text{Re } c > 0. \quad (7.9)$$

Quartic decay justifies holomorphic differentiation on compact subsets. For real $c > 0$, $(-1)^k \mathcal{C}^{(k)}(c) > 0$ for every $k \geq 0$, because it is the Gaussian expectation of $((A^4 + B^4 + A^2 B^2)/4)^k e^{-cq_4/4}$. In particular it is strictly decreasing and $\mathcal{C}(0+) = 1$.

Proposition 7.1 (Shape-period transport). *The period (7.9) satisfies*

$$48c^4 \mathcal{C}''' + (288c^3 + 28c^2) \mathcal{C}'' + (324c^2 + 56c + 4) \mathcal{C}' + (36c + 7) \mathcal{C} = 0. \quad (7.10)$$

Its first one-sided asymptotic derivatives are

$$\mathcal{C}(0+) = 1, \quad \mathcal{C}'(0+) = -7/4, \quad \mathcal{C}''(0+) = 297/16, \quad \mathcal{C}'''(0+) = -31815/64.$$

Proof. For the mass/coupling integral $I(M, g)$ of Section 4, radial integration by parts gives $(M\partial_M + 2g\partial_g)I = -I$. Writing $A_n(c) = \partial_M^n I(1, c)$ yields $A_{n+1} = -(n+1)A_n - 2cA'_n$, hence

$$\begin{aligned} A_1 &= -\mathcal{C} - 2c\mathcal{C}', \\ A_2 &= 2\mathcal{C} + 10c\mathcal{C}' + 4c^2\mathcal{C}'', \\ A_3 &= -6\mathcal{C} - 54c\mathcal{C}' - 48c^2\mathcal{C}'' - 8c^3\mathcal{C}'''. \end{aligned}$$

Insert these expressions into the fixed-quartic mass equation. The derivative values follow independently from Gaussian moments. Taylor's finite-order remainder for the exponential is integrably dominated on $\text{Re } c \geq 0$, so these are controlled asymptotic jets. No convergent Taylor series through the singular point $c = 0$ is asserted. $\square$

7.4 The physical-to-period ratio and scalar compensation

Let $H(c, \bar{c}) = g_{00}(c, \bar{c}) > 0$ be the raw identity-volume norm, and define $\mathcal{R} = H/|\mathcal{C}|^2$ where the period is nonzero. A common source rescaling cancels from this ratio.

Proposition 7.2 (No constant ratio on a complex-open shape region). *The ratio $\mathcal{R}$ is constant on no nonempty complex-open subset of $\text{Re } c > 0$. A common scalar source adjustment cannot preserve both a fixed identity Euler period and its physical Euler metric along a nonconstant holomorphic shape variation.*

Proof. If $H = K|\mathcal{C}|^2$ on such a patch, radially of H gives $\text{Im}(c\mathcal{C}'/\mathcal{C}) = 0$. This holomorphic expression is then a real constant k . The identity theorem extends $c\mathcal{C}' - k\mathcal{C} = 0$ throughout the right half-plane. Along the positive axis $\mathcal{C}(c) = Ac^k$; its limit one at zero forces $k = 0, A = 1$, contrary to strict decrease.

For a holomorphic shape $c = c(M)$, simultaneous preservation would require $a(M)\mathcal{C}(c(M)) = C_*$ and $|a(M)|^2 H(c(M), \bar{c}(M)) = H_*$ with $C_* \neq 0, H_* > 0$. Cancellation makes $\mathcal{R}$ constant on the image of $c(M)$, which is open unless $c(M)$ is constant. This proves the claim; the compensating scalar need not itself be holomorphic. $\square$

Even preservation of the identity period alone with a fixed scalar source forces $c(M)$ constant, by the open mapping theorem and nonconstancy of $\mathcal{C}$. A variable scalar $a(c) = \mathcal{C}(c_0)/\mathcal{C}(c)$ can restore that period, but the resulting physical metric retains the factor $\mathcal{R}$.

The angular discrepancy can be exhibited without evaluating H :

$$\frac{\mathcal{R}(re^{i\theta}, re^{-i\theta})}{\mathcal{R}(r, r)} = 1 - \frac{7}{2}r(1 - \cos \theta) + O(r^2), \quad |\theta| < \pi/2. \quad (7.11)$$

It follows from $\mathcal{C}(c) = 1 - 7c/4 + O(|c|^2)$ and exact radially. This does not establish variation or monotonicity of the ratio along the positive real axis alone. State mixing, different boundaries, cycles or target metrics change the hypotheses and remain possible.

8 Physical caps and the residue pairing

The raw real-cycle period is a useful holomorphic diagnostic, but it is not yet a functional on physical vacua. This section constructs such functionals and fixes their normalization. All norms use ordinary Euclidean measure on $\mathbb{C}^2$, with $\|dz_j\|^2 = 2$. We take Hermitian inner products to be antilinear in the first slot.

8.1 Compact representatives and long cylinders

For a strongly elliptic polynomial f , set

$$D_f = \bar{\partial} + df \wedge, \quad D_f^\dagger = 2 \sum_j \left(-\iota_{\bar{j}} \partial_{z_j} + \bar{f}_j \iota_j \right), \quad H_f = 2\{D_f, D_f^\dagger\}.$$

The realization is the closed Hilbert complex on square-integrable forms. Its finite-dimensional harmonic space is denoted $\mathcal{V}_f$. The polynomial models here satisfy the noncompact Hodge hypotheses; the degenerate homogeneous endpoint is checked in Section 9. The relevant comparison theorem does not require Morse critical points [11].

Choose a compactly supported smooth cutoff χ equal to one near every critical point. Off this set, define

$$V_f = \frac{(df \wedge)^\dagger}{|df|^2}, \quad N_f = [\bar{\partial}, V_f], \quad \mathcal{R}_f = V_f(1 + N_f)^{-1}.$$

Brackets here are graded. The inverse is a finite series by bidegree, and $[D_f, \mathcal{R}_f] = 1$. Consequently

$$T_\chi = 1 - [D_f, (1 - \chi)\mathcal{R}_f] = \chi + \bar{\partial}\chi \wedge \mathcal{R}_f, \quad [D_f, T_\chi] = 0. \quad (8.1)$$

For $\Omega = dz_1 \wedge dz_2$ and a polynomial p , the form $u_p = T_\chi(p\Omega)$ is compactly supported and closed. Its physical cap is

$$h_f(p) = P_f u_p = \lim_{t \rightarrow \infty} e^{-tH_f} u_p. \quad (8.2)$$

This is a state preparation by a long Euclidean cylinder. At finite length the gluing identity is $\langle e^{-tH_f} u_p, e^{-tH_f} u_q \rangle = \langle u_p, e^{-2tH_f} u_q \rangle$; the limiting pairing is the ordinary positive harmonic Gram matrix.

Proposition 8.1. *The cap in (8.2) is cutoff independent and depends only on the Jacobi class of $p\Omega$.*

Proof. Differences of the cutoff operators in (8.1) are D_f -commutators supported away from critical points, so their action on a closed polynomial top form is compactly supported and exact. If $p\Omega = df \wedge \gamma$ with γ a polynomial holomorphic one-form, then $T_\chi(p\Omega) = D_f(T_\chi \gamma)$ is also compactly supported and exact. Harmonic projection annihilates both kinds of difference. The Hodge–Jacobi comparison identifies the resulting classes with $\mathcal{V}_f$. $\square$

The same assertion fails for the unmodified real-cycle integral. For $f_c = G_c/4$ and $s = A^2 + B^2$, $q_4 = A^4 + B^4 + A^2 B^2$, let

$$z_c = Af_{c,A} + Bf_{c,B} = \frac{s + cq_4}{4}.$$

This is zero in the Jacobi ring. Real integration by parts nevertheless gives

$$\frac{1}{2\pi} \int_{\mathbb{R}^2} z_c e^{-G_c} dA dB = \frac{1}{2} \mathcal{C}(c) > 0 \quad (c > 0).$$

Indeed, integrating the divergence of $(A, B)e^{-G_c}$ gives $\int (s + cq_4)e^{-G_c} = 2 \int e^{-G_c}$. Thus the raw period cannot be identified with a physical cap functional on the same source classes.

8.2 Opposite twist and absolute normalization

Let $U = (-1)^{\text{holomorphic degree}}$. It unitarily intertwines f and $-f$. Hodge star followed by complex conjugation gives an antiunitary map $\mathfrak{C} : \mathcal{V}_{-f} \rightarrow \mathcal{V}_f$. In middle degree on $\mathbb{C}^2$, $\mathfrak{C}^2 = 1$ and $\mathfrak{C}U = U\mathfrak{C}$. Hence $\Theta = \mathfrak{C}U$ is an antiunitary involution of $\mathcal{V}_f$. The reflected caps and their functionals are

$$b_p = \Theta h_f(p), \quad \ell_p(q) = \langle b_p, h_f(q) \rangle = \int_{\mathbb{C}^2} h_f(q) \wedge h_{-f}(p). \quad (8.3)$$

The last pairing is complex bilinear. It is not a positive norm. Opposite twists are essential: for a form a of degree k ,

$$\int D_f a \wedge b = (-1)^{k+1} \int a \wedge D_{-f} b.$$

This identity supplies cohomological descent of the wedge pairing.

The proportionality to the residue pairing is fixed by a Gaussian, not chosen to match an Euler factor. In one coordinate, for $f = az^2/2$ and $a \neq 0$, representatives of dz are

$$\alpha_{\pm} = e^{-|a||z|^2} \left(dz \mp \frac{\bar{a}}{|a|} d\bar{z} \right).$$

Their exact class normalization follows by applying $D_{\pm f}$ to $(e^{-|a||z|^2} - 1)/(\pm az)$. Direct integration gives $\|\alpha_{\pm}\|^2 = 2\pi/|a|$ and $\int \alpha_+ \wedge \alpha_- = -2\pi i/a$. The additional reordering sign for two coordinates yields $4\pi^2/(ab)$ for a diagonal Hessian (a, b) . With the residue orientation fixed by Ω , localization and deformation of the cohomological pairing therefore give

$$\ell_p(q) = 4\pi^2 \text{Res}_f(pq). \quad (8.4)$$

The opposite-twist Hodge pairing underlying this calculation is discussed in [11]; the displayed constant is specific to the Euclidean and differential conventions used here.

In the shape frame E of (7.4), put

$$\lambda = \frac{256\pi^2}{3}, \quad P = 4\pi^2 \eta_E = \lambda \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & -4/3 \\ 1 & -4/3 & 4/9 \end{pmatrix}.$$

If $G_{ij} = \langle h_i, h_j \rangle$ is the physical Gram matrix and $\mathcal{B}_{ij} = \langle b_i, b_j \rangle$ is the cap Gram matrix, then

$$\mathcal{B} = PG^{-1}P^{\dagger} = G^T = \bar{G}. \quad (8.5)$$

The first equality is the Riesz representation formula; the second uses the antiunitary involution. In particular, residue data alone do not determine G .

8.3 The exact boundary Gram identity

For any positive finite-dimensional metric G and genuine boundary map $v \mapsto y = Pv$, let $\mathcal{B} = PG^{-1}P^{\dagger}$. On the range of P , the minimum-norm state with boundary values y is $v_{\min} = G^{-1}P^{\dagger}\mathcal{B}^+y$, where $+$ denotes the Moore–Penrose inverse. Orthogonal decomposition gives

$$\|v\|_G^2 = y^{\dagger}\mathcal{B}^+y + \|v - v_{\min}\|_G^2. \quad (8.6)$$

For one functional $\ell(v) = pv$ this reads $\|v\|_G^2 = |\ell(v)|^2/(pG^{-1}p^{\dagger}) + \|v_{\perp}\|_G^2$. For invertible P all states are detected and $G = P^{\dagger}\mathcal{B}^{-1}P$. This is the physical replacement for equating one raw period to one norm. For example, $\ell_1(1) = 0$ although $\|h_f(1)\|^2 > 0$; the top cap detects the identity through $\ell_v(1) = \lambda$.

8.4 A normalized homogeneous prime source

At $f_0 = (Z^4 + W^4 + Z^2W^2)/16$, rotation symmetry makes the physical metric in $(1, s_0, t_0)\Omega_Z$, where $s_0 = Z^2 + W^2$ and $t_0 = Z^2W^2$, diagonal. Its residue matrix is λ times the anti-diagonal identity. Equation (8.5) then gives

$$G_0 = \text{diag}(a_0, \lambda, \lambda^2/a_0), \quad a_0 > 0. \quad (8.7)$$

The middle class is self-reflected after normalization: $\chi = h_{f_0}(s_0)/\sqrt{\lambda}$ has norm and bilinear self-pairing equal to one.

For the homogeneous prime model $f_M = M^2(U^4 + V^4 + U^2V^2)/16$, $M \neq 0$, the change of fields $\phi_M(U, V) = (\sqrt{M}U, \sqrt{M}V)$ gives the source

$$\sigma_M = \frac{\sqrt{3}M}{16\pi}(U^2 + V^2) dU \wedge dV = \frac{1}{M\sqrt{\lambda}}\phi_M^*(s_0\Omega_Z). \quad (8.8)$$

Degree-two Euclidean norms are invariant under the relevant dilation. Thus its harmonic norm is $|M|^{-2}$ and its bilinear self-pairing is M^{-2} . Taking $M = 1 - q$ gives a normalized interacting Euler metric. This uses the middle Jacobi class, not the identity class. The mass direction is still a dilation and remains zero in the Jacobi ring.

9 Endpoint geometry and conditional uniqueness

The physical caps give useful boundary data for the shape metric. One endpoint coalesces the critical points; the other sends eight vacua to infinity in the original coordinates. Both must be treated in a fixed source normalization.

9.1 The homogeneous endpoint

Write

$$f_\epsilon(Z, W) = \frac{Z^4 + W^4 + Z^2W^2}{16} + \frac{\epsilon}{8}(Z^2 + W^2), \quad \epsilon = c^{-1/2}.$$

This follows from $Z = c^{1/4}A$, $W = c^{1/4}B$. The leading cubic gradient has no zero on the unit sphere: its equations are $Z(2Z^2 + W^2) = W(2W^2 + Z^2) = 0$, whose only common solution is zero. Uniformly for bounded ϵ , with $R^2 = |Z|^2 + |W|^2$,

$$|\nabla f_\epsilon| \geq aR^3 - bR, \quad |\nabla^k f_\epsilon| \leq C_k(1 + R^{4-k}), \quad k \geq 2.$$

It follows that for every $\delta > 0$ and relevant k , $\delta|\nabla f_\epsilon|^k - |\nabla^k f_\epsilon| \rightarrow +\infty$. Euclidean space has bounded geometry, so these estimates establish the strong ellipticity hypotheses of the non-Morse Hodge comparison theorem [11, Theorem 2.34].

Proposition 9.1. *The harmonic spaces of f_ϵ , including $\epsilon = 0$, have dimension nine. Their projections and the Gram matrices in fixed compact cohomology representatives vary continuously at zero.*

Proof. The two leading cubics form a regular sequence. Their quotient has Hilbert series $(1 + z + z^2)^2$ and dimension nine. Lower-degree terms preserve this dimension, and Hodge comparison identifies it with the middle-degree kernel; other degrees vanish. The quadratic forms of the Hamiltonians have common domain $H^1 \cap L^2(R^6 dV)$ on forms. The variation in their coefficients is bounded by $C(|\epsilon|R^4 + |\epsilon|^2R^2 + |\epsilon|)$, which is small relative to the common confining form as $\epsilon \rightarrow 0$. Form continuity gives norm-resolvent continuity. Compact resolvent and the constant nine-dimensional kernel give a local spectral gap and continuous kernel projections. A single cutoff enclosing all critical points in this neighborhood makes the homotopy representatives in (8.1) continuous as well. $\square$

At zero, combining field rotation with target rotation gives characters $e^{i(d-6)\theta}$ on the polynomial-degree d classes. The degrees 0, 2, 4 in the determinant-character sector are distinct, so its Gram matrix is diagonal. Combining continuity with (8.7), the radial shape metric satisfies

$$g_E(\rho) = D_\rho \left[\text{diag}(a_0, \lambda, \lambda^2/a_0) + o(1) \right] D_\rho, \quad D_\rho = \text{diag}(\sqrt{\rho}, 1, \rho^{-1/2}), \quad \rho = 1/c \downarrow 0. \quad (9.1)$$

The assertion concerns the rescaled matrix. It does not assert that every unscaled off-diagonal entry tends to zero. For the identity class this gives $H(c, c) \sim a_0/c$. Direct real scaling gives $\mathcal{C}(c) \sim b_0/\sqrt{c}$, where

$$b_0 = \frac{1}{2\pi} \int_{\mathbb{R}^2} e^{-(Z^4+W^4+Z^2W^2)/4} dZ dW > 0.$$

Consequently $H(c, c)/\mathcal{C}(c)^2 \rightarrow a_0/b_0^2$; no value of a_0 has been inferred from the raw period.

9.2 The massive endpoint without a uniform gap assumption

For $c > 0$, use $Z = \sqrt{c}A$, $W = \sqrt{c}B$ instead. Then $f_c = \phi_c^*(\rho F)$ with $\rho = 1/c$ and $F = f_1$. The frame obeys $E = \rho \phi_c^*((1, S, T)\Omega_Z)$, so $g_E = \rho^2 g_{\text{ref}}$. All nine critical points of F are now fixed and Morse.

Let H_v be the complex Hessian at a critical point. A unitary Takagi factorization gives positive singular values and local holomorphic Morse coordinates with unit-modulus Jacobian k_v . The product Gaussians of Section 8, corrected by the compact homotopy outside disjoint small balls, give exact-class representatives for the primitive idempotents at twists $\pm \rho F$. Their supports are disjoint and their squared norms satisfy

$$N_v^\pm = \frac{4\pi^2}{\rho^2 |\det H_v|} (1 + O(\rho^{-1/2})).$$

The relative error follows from the first variation of the Euclidean metric in Morse coordinates on the scale $|\zeta| \asymp \rho^{-1/2}$; cutoff corrections are exponentially small. Their opposite-twist pairing is exactly diagonal, with entries $k_{v,\rho} = 4\pi^2/(\rho^2 \det H_v)$.

Proposition 9.2. *In the primitive-idempotent frame,*

$$\rho^2 G_{\text{ref}} = 4\pi^2 \text{diag}_v(|\det H_v|^{-1}) + O(\rho^{-1/2}).$$

Proof. Harmonic representatives minimize norm in their class, giving $G_{\text{ref}} \leq \text{diag } N_v^+$. For a source with coefficients z_v , pair with the compact opposite-twist representative of coefficients $\overline{z_v k_{v,\rho}}/N_v^-$. Cauchy–Schwarz and exact descent give the lower matrix bound $\text{diag}(|k_{v,\rho}|^2/N_v^-) \leq G_{\text{ref}}$. The two bounds have the same leading term. $\square$

This proof needs no spectral gap uniform in ρ . It establishes the metric limit, not a derivative expansion or Stokes data.

The three orbit sums of $|\det H_v|^{-1}$ are 16, 64, 48. The evaluation matrix from the polynomial frame to these orbits is

$$\mathbf{V} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 1 & -4/3 & 4/9 \end{pmatrix}.$$

Thus the other endpoint is completely fixed:

$$g_E(\rho) \longrightarrow g_\infty = 4\pi^2 \begin{pmatrix} 128 & -128 & 64/3 \\ -128 & 448/3 & -256/9 \\ 64/3 & -256/9 & 256/27 \end{pmatrix}, \quad \rho \rightarrow \infty. \quad (9.2)$$

In particular, $H(c, c) = 512\pi^2 + O(\sqrt{c})$ as $c \downarrow 0$. The exactly quadratic theory at $c = 0$ has identity norm $64\pi^2$. The difference is explained by the two escaping orbits, whose limiting contributions are $256\pi^2$ and $192\pi^2$. It is not legitimate to replace the small-positive- c nine-vacuum limit by the one-vacuum Gaussian model.

9.3 Uniqueness under the physical chiral equation

The following statement is conditional on the exact identification of the physical connection and chiral equation in (7.7). Endpoint estimates alone do not establish that identification.

Theorem 9.3. *For the same Higgs matrix $C = R/16$, there is at most one positive C^2 metric on the punctured shape plane satisfying (7.7), (9.1), and (9.2). The constant $a_0 > 0$ in the first endpoint need not be prescribed. For nonradial solutions, the endpoint conditions are required uniformly on circles.*

Proof. For two metrics put $F_{12} = \text{tr}(g_1^{-1}g_2)$. At a point choose a holomorphic normal frame for g_1 , then a constant unitary change with $g_2 = \Lambda = \text{diag}(\lambda_i)$. Writing $X = \partial g_2$, direct substitution of the two metric equations gives

$$\bar{\partial}\partial F_{12} = \left\| \Lambda^{-1/2} X \right\|_{\text{HS}}^2 + \sum_{i,j} \frac{(\lambda_i - \lambda_j)^2}{\lambda_j} |C_{ij}|^2 \geq 0.$$

Therefore $d = F_{12} + F_{21} - 6$ is nonnegative and subharmonic. At the small endpoint its limit is $2(a_2/a_1 + a_1/a_2 - 2)$, where a_i are the two possible constants, so it is bounded. At infinity it tends to zero because both metrics tend to g_∞ . On $\epsilon < |u| < R$, bound d by the harmonic function

$$\delta_R + (K - \delta_R) \frac{\log(R/|u|)}{\log(R/\epsilon)},$$

where K bounds the inner boundary and $\delta_R = \sup_{|u|=R} d$. Let $\epsilon \downarrow 0$ and then $R \rightarrow \infty$. This proves $d = 0$ and hence $g_1 = g_2$. $\square$

The physical harmonic metric exists throughout the punctured parameter plane by the local uniform form estimates, the constant-dimensional kernel and fixed compact representatives. If it satisfies the stated chiral equation in this frame, it supplies existence as well as uniqueness. In that case the massive endpoint selects a_0 implicitly. Neither its numerical value nor a new explicit solution of the nonlinear radial equation is asserted.

10 The closed arithmetic source and its spectral scale

Finite-dimensional ground-state geometry does not supply the full arithmetic input space. We now construct a closed infinite-dimensional preparation for the gamma kinetic energy, retaining all remaining arithmetic terms explicitly.

10.1 Local fields, maximal domain and adjoint

Let $H = -\partial_x^2$ on $L^2(\mathbb{R})$, $a_k = 2k + 1/2$, and $R_k = (a_k^2 + H)^{-1}$. For a prescribed source F , minimize

$$\mathcal{E}_k(F, u) = \frac{2}{a_k} \left(\|F - u\|^2 + a_k^{-2} \|u'\|^2 \right) \quad \text{over } u \in H^1(\mathbb{R}).$$

The minimizer is $u_k = a_k^2 R_k F$. Define the output space and residuals

$$\mathcal{Y}_\gamma = \bigoplus_{k \geq 0} (L^2(\mathbb{R}) \oplus L^2(\mathbb{R})), \quad (\mathcal{A}_\gamma F)_k = \begin{pmatrix} \sqrt{2/a_k} H R_k F \\ \sqrt{2a_k} \partial_x R_k F \end{pmatrix}. \quad (10.1)$$

The Fourier column $v_k(\tau)$ obtained from this expression has $v_k^\dagger v_k = (2/a_k)\tau^2/(a_k^2 + \tau^2)$. Hence

$$\|\mathcal{A}_\gamma F\|^2 = \frac{1}{2\pi} \int_{\mathbb{R}} B(\tau^2) |\widehat{F}(\tau)|^2 d\tau = K[F].$$

The digamma partial fraction and asymptotic formulas [12] give

$$B(\tau^2) = \operatorname{Re} \psi(1/4 + i\tau/2) - \psi(1/4) = \log |\tau| - \log 2 - \psi(1/4) + O(\tau^{-2}). \quad (10.2)$$

The positive measure $\sum_k (2/a_k) \delta_{a_k^2}$ also places this explicit tower in the complete-Bernstein extension framework [13].

For $X_L = L^2(I_L)$ and zero extension E_L , write $A_L = \mathcal{A}_\gamma E_L$ and

$$\mathcal{D}_{\log, L} = \left\{ f \in X_L : \int_{\mathbb{R}} B(\tau^2) |\widehat{E_L f}(\tau)|^2 d\tau < \infty \right\}. \quad (10.3)$$

This is the maximal source domain. It is equivalent to the Fourier weight $1 + \log(2 + |\tau|)$ after adding the L^2 norm.

Proposition 10.1. *The source A_L is closed and densely defined, with domain $\mathcal{D}_{\log, L}$ and core $C_c^\infty(I_L)$.*

Proof. Each component of (10.1) is bounded. If $f_n \rightarrow f$ and $A_L f_n \rightarrow y$ in the two Hilbert spaces, componentwise convergence identifies every component of y with that of $A_L f$. Their squared sum is finite, proving closedness. For the core assertion, first contract the supported function using $D_r f(x) = r^{-1/2} f(x/r)$, $r < 1$. These maps are uniformly bounded near $r = 1$ in the logarithmic Fourier norm and converge strongly to the identity there, by density and the weight comparison under dilation. The contracted support has a positive gap from the endpoints. Mollification inside that gap then converges in the same norm. $\square$

The maximal adjoint must be taken after interval restriction. For $\Psi \in \mathcal{Y}_\gamma$, define the distribution t_Ψ by

$$\widehat{t_\Psi}(\tau) = \sum_k v_k(\tau)^\dagger \widehat{\Psi}_k(\tau).$$

Cauchy–Schwarz gives $|\widehat{t_\Psi}|^2/(1+B) \leq \sum_k |\widehat{\Psi}_k|^2$, so the sum defines a distribution. The global adjoint requires $t_\Psi \in L^2(\mathbb{R})$, whereas

$$\operatorname{Dom} A_L^* = \{\Psi : t_\Psi|_{I_L} \in L^2(I_L)\}, \quad A_L^* \Psi = t_\Psi|_{I_L}. \quad (10.4)$$

In particular the interval adjoint domain can be larger. The form operator $T_L = A_L^* A_L$ has domain

$$\operatorname{Dom} T_L = \{f \in \mathcal{D}_{\log, L} : (B(H)E_L f)|_{I_L} \in L^2(I_L)\}, \quad (10.5)$$

where the multiplier is first understood distributionally. The block charge with entries A_L, A_L^* on $\operatorname{Dom} A_L \oplus \operatorname{Dom} A_L^*$ is self-adjoint, and its square has diagonal entries $T_L, A_L A_L^*$. This construction concerns the kinetic term.

10.2 Two controlled singular limits

Let $A_{L,N}$ retain $0 \leq k < N$. Its exact operator norm is

$$\|A_{L,N}\|^2 = \sum_{k < N} \frac{2}{a_k} = \psi(N + 1/4) - \psi(1/4) = \log N - \psi(1/4) + O(N^{-1}). \quad (10.6)$$

The upper bound follows from the multiplier; modulating any fixed nonzero compactly supported input to arbitrarily high frequency attains its supremum, including after compression. Increasing towers converge in output norm precisely on $\mathcal{D}_{\log,L}$; outside it their squared norms diverge. If $F = E_L f \in H^1(\mathbb{R})$, the tail obeys

$$\|(A_L - A_{L,N})f\|^2 \leq \left(\frac{2}{a_N^3} + \frac{1}{2a_N^2} \right) \|F'\|^2,$$

where $A_{L,N}$ is embedded by zero components. Monotone convergence of closed forms gives strong-resolvent convergence of the associated operators [14].

The heat regulator is $A_{L,\epsilon} = \mathcal{A}_\gamma e^{-\epsilon H/2} E_L$. Its point-source columns are the inverse Fourier transforms of $v_k(\tau) e^{-\epsilon \tau^2/2} e^{-i\tau x}$, and

$$\begin{aligned} \|A_{L,\epsilon}\|_{\text{HS}}^2 &= \frac{L}{2\pi} \int_{\mathbb{R}} B(\tau^2) e^{-\epsilon \tau^2} d\tau \\ &= \frac{L}{2\sqrt{\pi\epsilon}} \left(\frac{1}{2} \log \frac{1}{\epsilon} + \frac{\gamma_E}{2} + \frac{\pi}{2} + \log 2 \right) + O_L(1). \end{aligned} \quad (10.7)$$

To obtain the constant, integrate the logarithm in (10.2), using $\psi(1/2) = -\gamma_E - 2\log 2$ and $\psi(1/4) = -\gamma_E - \pi/2 - 3\log 2$. The remainder is integrable after subtracting the high-frequency asymptotic with a cutoff. The source and form limits again have exactly domain $\mathcal{D}_{\log,L}$, with strong-resolvent convergence. Thus a regularized family of normalizable point caps has a necessarily singular size limit.

10.3 The complete signed form

Let $c(x) = \cosh(x/2)$, $s(x) = \sinh(x/2)$ on I_L , and $T_d = E_L^* U_d E_L$. Define the bounded self-adjoint remainder

$$R_L = w_0 I + 2cc^* - 2ss^* - \sum_{m \log p < L} (\log p) p^{-m/2} (T_{m \log p} + T_{m \log p}^*). \quad (10.8)$$

Here cc^* denotes $f \mapsto c \langle c, f \rangle$; the squared integrals in the original convention have the same quadratic values. A useful coarse bound is

$$\|R_L\| \leq |w_0| + L + 2 \sinh(L/2) + 2 \sum_{m \log p < L} (\log p) p^{-m/2}.$$

The complete operator is $W_L = T_L + R_L$, with form domain $\mathcal{D}_{\log,L}$ and operator domain $\text{Dom } T_L$. Adding this same R_L to either regulator preserves the complete signed expression and the strong-resolvent limit. These operators are uniformly bounded below; that fact is not their nonnegativity. If $I_L \subset I_{L^*}$, zero extension preserves the complete quadratic form, since all newly available translations with no overlap vanish.

The embedding $\mathcal{D}_{\log,L} \hookrightarrow X_L$ is compact. Indeed, the weighted norm uniformly controls the high-frequency tail, while restricting a fixed finite Fourier band to a finite interval is Hilbert–Schmidt. Thus T_L and W_L have compact resolvent. There are only finitely many negative eigenvalues of W_L at any fixed L , without a claim about their absence.

10.4 Why bounded and finite-rank sources do not suffice

For $\varphi \in C_c^\infty(I_L)$ and $f_N(x) = e^{iNx}\varphi(x)$, expansion of the multiplier gives

$$Q_L[f_N] = \|\varphi\|^2 (\log N - \log(2\pi)) - 2 \sum_{m \log p < L} (\log p) p^{-m/2} \operatorname{Re} \left(e^{-iNm \log p} \langle E_L \varphi, U_{m \log p} E_L \varphi \rangle \right) + o(1). \quad (10.9)$$

The pole amplitudes decay faster than any power. In particular $Q_L[f_N] = \|\varphi\|^2 \log N + O(1)$. No bounded source can represent this form. A square-integrable family of point caps b_x would define a Hilbert–Schmidt source and is also excluded. Evolution confined to a finite vacuum space, on which $H = 0$, does not supply these inputs.

Nor can an unbounded finite-range source evade the obstruction by a domain restriction. Every densely defined closable operator with finite-dimensional range is bounded on its domain: otherwise choose inputs tending to zero whose outputs have norm one, then take a convergent output subsequence, contradicting closability. Its closure is therefore bounded and everywhere defined. If a proposed norm identity holds on the smooth core, the closed arithmetic form makes that source closable: a sequence of inputs tending to zero with Cauchy outputs is Cauchy in the shifted form norm, so the closed-form limit forces the output limit to vanish. Equation (10.9) excludes the resulting bounded operator.

10.5 A resolvent pairing as a general control

For an independently self-adjoint physical Hamiltonian $\mathcal{H}$ and bounded finite-channel coupling J , put $M(z) = J^*(\mathcal{H} - z)^{-1}J$ for $\operatorname{Im} z > 0$ and $\Gamma(z) = (\mathcal{H} - \bar{z})^{-1}J$. The resolvent identity gives

$$\frac{M(z) - M(w)^*}{z - \bar{w}} = \Gamma(z)^* \Gamma(w). \quad (10.10)$$

This is a positive Gram kernel. Its Cayley transform $S(z) = (M(z) - iI)(M(z) + iI)^{-1}$ satisfies

$$\frac{I - S(z)S(w)^*}{-i(z - \bar{w})} = 2(M(z) + iI)^{-1} \frac{M(z) - M(w)^*}{z - \bar{w}} (M(w)^* - iI)^{-1}.$$

In a general operator Herglotz representation

$$M(z) = A + Dz + \int_{\mathbb{R}} \left(\frac{1}{t - z} - \frac{t}{1 + t^2} \right) d\Sigma(t), \quad D \geq 0,$$

the kernel sees D and the positive measure but loses the constant A [15]. This illustrates both the strength and the normalization limit of a resolvent Gram law. A regular bounded coupling does not on its own identify the complete arithmetic form or its singular source limit.

10.6 Neumann–Dirichlet comparison

Let H_N, H_D be the ordinary Neumann and Dirichlet interval Laplacians.

Theorem 10.2. *In the sense of closed quadratic forms,*

$$B(H_N) \leq T_L \leq B(H_D),$$

with $\operatorname{Dom} B(H_D)^{1/2} \subset \mathcal{D}_{\log, L} \subset \operatorname{Dom} B(H_N)^{1/2}$. Consequently, indexing from one,

$$B(((n-1)\pi/L)^2) \leq \lambda_n(T_L) \leq B((n\pi/L)^2), \quad \lambda_n(T_L) = \log \frac{\pi n}{2L} - \psi(1/4) + O_L(n^{-1}). \quad (10.11)$$

Proof. For each mass, restricting a whole-line auxiliary field to I_L drops nonnegative exterior energy; extending a Dirichlet field by zero gives an admissible whole-line competitor. Minimizing the three energies therefore orders them as Neumann, whole-line, Dirichlet. The outer minima are the forms of $(2/a)H_{N,D}(a^2 + H_{N,D})^{-1}$. Sum finite towers and pass to increasing form limits. Min-max and the known interval Laplacian eigenvalues give the bounds; (10.2) gives the asymptotic. $\square$

This comparison concerns three different preparations. It does not impose Neumann or Dirichlet boundary conditions on the actual whole-line gamma fields.

11 Pole gluing, Robin response and coherent delays

There are two explicit ways to produce the pole kernel $P_L(x, y) = 2 \cosh((x - y)/2)$. The first changes a boundary response and exposes an unstable auxiliary mode. The second interferes with remote delayed sources and adds a positive contact.

11.1 The attractive Robin identity

Put $a = 1/2$, $h = L/2$, and let R_f be the interval integral operator with kernel $e^{-|x-y|/2}$. The lowest gamma contribution is $K_0 = 4I - R_f$. Define

$$S_R u = -u'', \quad A_R = S_R + a^2,$$

$$\text{Dom } S_R = \{u \in H^2(-h, h) : u'(h) = au(h), u'(-h) = -au(-h)\}.$$

Its closed form on $H^1(-h, h)$ is

$$\mathfrak{a}_R[u] = \int_{-h}^h (|u'|^2 + a^2|u|^2) dx - a(|u(h)|^2 + |u(-h)|^2).$$

Endpoint trace bounds prove semiboundedness and closedness. Direct differentiation, the derivative jump and the Robin conditions give

$$A_R^{-1}(x, y) = -e^{|x-y|/2} = R_f(x, y) - P_L(x, y), \quad C_L := K_0 + P_L = 4I - A_R^{-1}. \quad (11.1)$$

There is no zero mode of A_R .

The free whole-line auxiliary field, eliminated outside the interval, has energy $4\|f - u\|^2 + 16\|u'\|^2 + 8(|u(h)|^2 + |u(-h)|^2)$. Producing the pole law reverses the sign of the endpoint term. Equivalently the resulting energy is

$$\mathcal{E}_R(f, u) = 4\|f\|^2 - 8\text{Re}\langle f, u \rangle + 16\mathfrak{a}_R[u].$$

Its stationary point is $u_f = A_R^{-1}f/4$, and

$$\mathcal{E}_R(f, u) = \langle f, C_L f \rangle + 16\mathfrak{a}_R[u - u_f]. \quad (11.2)$$

This is a stationary elimination, not a nonnegative minimization.

Indeed A_R has exactly one negative eigenvalue $-\delta_e$, where

$$\kappa_e \tanh(\kappa_e h) = a, \quad \delta_e = \kappa_e^2 - a^2 > 0, \quad \phi_e(x) \propto \cosh(\kappa_e x).$$

The even equation has a unique root $\kappa_e > a$. An odd negative eigenvalue of A_R would require $\kappa \coth(\kappa h) = a$ with $\kappa > a$, which is impossible. Thus $\inf_u \mathcal{E}_R(f, u) = -\infty$ for every f .

Proposition 11.1. *The bounded reduced operator C_L is strictly positive for $L < 4$, has kernel spanned by x at $L = 4$, and has exactly one negative eigenvalue, in the odd sector, for $L > 4$.*

Proof. On an eigenvector of S_R with eigenvalue ν , the coefficient of C_L is $4\nu/(a^2 + \nu)$. The even negative eigenvalue has $\nu < -a^2$ and gives a positive coefficient. An odd negative eigenvalue of S_R exists exactly when $ah > 1$, from $\kappa_o \coth(\kappa_o h) = a$, $0 < \kappa_o < a$. It gives the single negative coefficient. At $ah = 1$, the odd zero mode is x . The remaining eigenvalues are positive. As direct checks, $C_L 1 = 4e^{L/4} \cosh(x/2)$ and $C_L x = 8e^{L/4}(1 - L/4) \sinh(x/2)$. $\square$

For $L > 4$ write $r_L = (a^2 - \kappa_o^2)^{-1} - 4 > 0$ and let Π_o project onto the normalized odd mode. Set $r_L \Pi_o = 0$ for $L \leq 4$. Removing the negative even auxiliary direction makes (11.2) a minimization, but its reduced operator is $4I - A_R^{-1}(I - \Pi_e)$; recovering C_L adds $\delta_e^{-1} \Pi_e$. This does not remove the odd reduced defect.

There is also an explicit positive source for $C_L + r_L \Pi_o$. Let S_+ be the nonnegative part of S_R and $P_{\geq 0}$ its spectral projection. Use

$$\mathcal{B}_L f = \begin{pmatrix} 2S_+(S_+ + 1/4)^{-1} P_{\geq 0} f \\ S_+^{1/2}(S_+ + 1/4)^{-1} P_{\geq 0} f \\ \sqrt{4 + \delta_e^{-1}} \langle \phi_e, f \rangle \end{pmatrix}.$$

Functional calculus gives $\mathcal{B}_L^* \mathcal{B}_L = C_L + r_L \Pi_o$. All components are bounded; the adjoint is the sum of the transposed components with conjugated scalar coefficients. Adjoining the higher gamma modes yields a closed source of norm $K + P_L + r_L \Pi_o$. The full arithmetic form still has the contact, translations and $-r_L \Pi_o$ to account for.

11.2 A supersymmetric factorization needs a shift

With $w(x) = \kappa_e \tanh(\kappa_e x)$, take $D = \partial_x - w$ on H^1 and $D^* = -\partial_x - w$ on H_0^1 . Then

$$D^* D = S_R + \kappa_e^2 = A_R + \delta_e, \quad DD^* = -\partial_x^2 + \kappa_e^2 - 2\kappa_e^2 \operatorname{sech}^2(\kappa_e x)$$

with Dirichlet partner conditions. This is consistent with the boundary-domain requirements in supersymmetric quantum mechanics [16]. Using $D^* D$ as the kinetic operator in the resolvent changes the reduced operator to $C_{\text{shift}} = 4I - (A_R + \kappa_e^2)^{-1}$. Its difference from the desired one is

$$C_L - C_{\text{shift}} = -\kappa_e^2 A_R^{-1} (A_R + \kappa_e^2)^{-1}.$$

It is infinite rank. Positive factorization of a shifted Hamiltonian does not restore the unshifted pole identity.

11.3 Remote interference with a gamma channel

Fix a mass a and put $\alpha_a = \sqrt{2/a}$. Its gamma column is $\alpha_a((I - a^2 R_a)F, a\partial_x R_a F)$. Tensoring with the unit quartic cap χ in Section 8 leaves all the following Gram identities unchanged. Add to this same output channel

$$\mathcal{B}_{D,t} F = -t \begin{pmatrix} (U_D + U_{-D})F \\ (U_D - U_{-D})F \end{pmatrix}, \quad t \in \mathbb{R}.$$

Its square is $4t^2 I$. Writing $V = U_D + U_{-D}$ and $J = U_D - U_{-D}$, its cross term with the gamma source is

$$-2\alpha_a t V + 2\alpha_a t a^2 R_a V + 2\alpha_a a t \partial_x R_a J.$$

The multiplier is $-4\alpha_a t(\tau^2 \cos(D\tau) - a\tau \sin(D\tau))/(a^2 + \tau^2)$. For $D \geq L$ the atoms disappear after interval compression and the remaining kernel is $4\alpha_a t e^{-aD} \cosh(a(x-y))$. Taking $a = 1/2$ and $t = e^{D/2}/2$ gives a closed source A_0 with

$$T_0 := A_0^* A_0 = T_L + P_L + e^D I, \quad T_0 \geq mI, \quad m = e^D + L - 2 \sinh(L/2) > 0. \quad (11.3)$$

The lower bound drops the nonnegative gamma and even pole components and uses $\|s\|^2 = \sinh(L/2) - L/2$. Here and below $D \geq L$ is fixed.

The contact e^D is optimal in the two-remote-shift ansatz with arbitrary constant internal cap vectors. If $r_\pm, s_\pm$ are their scalar overlaps with the two gamma residuals, the pole coefficient $\kappa \cosh(a(x-y))$ fixes $X = r_+ + \bar{r}_- + s_+ - \bar{s}_- = -\kappa e^{aD}/(\alpha_a a)$. Cauchy-Schwarz bounds the contact below by

$$|X|^2/4 = \kappa^2 e^{2aD}/(8a).$$

For the pole normalization this is e^D , attained above. The statement is restricted to this ansatz.

11.4 Finite delay rigidity

A one-sided prime injection $-(c/\alpha_a)(U_d F, 0)$ has square plus cross term

$$\frac{ac^2}{2} I - c(U_d + U_{-d}) + ca^2 R_a(U_d + U_{-d}).$$

The final term is a continuous kernel $ca(e^{-a|x-y-d|} + e^{-a|x-y+d|})/2$. Thus an isolated shifted source produces the desired atoms along with a continuous distortion and a contact. Several shifts also produce autocorrelations between their outputs.

More generally let the delay alphabet $\mathcal{D}$ be finite and let the cap vectors in channel k and residual component $\varepsilon = 0, 1$ obey

$$\sum_{k,\varepsilon} \left(\sum_{d \in \mathcal{D}} \|z_{k,\varepsilon,d}\| \right)^2 < \infty.$$

This defines a bounded finite-delay source $\mathcal{B}$. Its zero-delay autocorrelation coefficient is the nonnegative $\gamma_{\mathcal{B}} = \sum_{k,\varepsilon,d} \|z_{k,\varepsilon,d}\|^2$.

Proposition 11.2. *Within this class, if the continuous gamma kernel is preserved (or changed only by the specified pole kernel), every added interior atom comes from $\mathcal{B}^* \mathcal{B}$. The contact coefficient is $\gamma_{\mathcal{B}}$. The cross term with the gamma source cannot independently supply nonzero interior prime atoms while its continuous part is cancelled.*

Proof. Let $u_{k,d}, w_{k,d}$ be the overlaps of the cap vectors with the two gamma residuals, and put $\sigma_{k,d} = u_{k,d} + \bar{u}_{k,-d}$, $\theta_{k,d} = \bar{w}_{k,-d} - w_{k,d}$. The cross term in channel k is

$$\alpha_{a_k} \sum_d \sigma_{k,d} U_d - \alpha_{a_k} a_k^2 R_k \sum_d \sigma_{k,d} U_d + \alpha_{a_k} a_k \partial_x R_k \sum_d \theta_{k,d} U_d.$$

Away from the finitely many centers, its continuous kernel is

$$-\sum_{k,d} \frac{\alpha_{a_k} a_k}{2} [\sigma_{k,d} + \theta_{k,d} \operatorname{sgn}(z-d)] e^{-a_k |z-d|}.$$

On each interval between centers this is an absolutely convergent Laurent expansion in $\zeta = e^{z/2}$ with powers $\pm(4k+1)$. Uniqueness of Laurent coefficients identifies it with zero, or with the two prescribed pole powers. Comparing adjacent intervals forces $\sigma_{k,d} = \theta_{k,d} = 0$ at every interior center. Remote centers are smooth across the interval and supply no interior atoms. The source autocorrelation is therefore the only remaining atomic contribution, with the stated zero coefficient. $\square$

Filters, resolvent feedback, and infinite alphabets with accumulating delays are outside this result. These distinctions matter in the constructions below.

12 Prime returns from a positive graph

Repeated prime shifts can be generated by a unitary return channel. The construction fixes their full geometric series and displays the contact required by positive output power.

12.1 A nonnegative graph Hamiltonian and its scattering matrix

For a prime p , set $q = p^{-1/2}$ and $d = \log p$. Use two half-line leads and a finite stub of length $d/2$ with a Neumann endpoint. At the common vertex choose the real symmetric orthogonal matrix

$$O_q = \begin{pmatrix} 1-q & \sqrt{q} & \sqrt{q(1-q)} \\ \sqrt{q} & 0 & -\sqrt{1-q} \\ \sqrt{q(1-q)} & -\sqrt{1-q} & q \end{pmatrix}, \quad P_{\pm} = (I \pm O_q)/2.$$

The quadratic form $\sum_{\text{edges}} \int |\psi'|^2$ on edgewise H^1 functions with $P_- \psi(0) = 0$ is closed and non-negative. Its self-adjoint Laplacian has vertex conditions $P_- \psi(0) = 0$, $P_+ \psi'(0) = 0$, with outgoing derivatives. This is the standard self-adjoint graph-scattering framework [17], specialized to an explicit coupler.

Write O_q in blocks $\begin{pmatrix} D_q & B_q^T \\ B_q & q \end{pmatrix}$, where $D_q = \begin{pmatrix} 1-q & \sqrt{q} \\ \sqrt{q} & 0 \end{pmatrix}$ and $B_q = \sqrt{1-q}(\sqrt{q}, -1)$. The stub round trip at wave number k multiplies by $z = e^{ikd}$. For lead input a , stub amplitude v and output b , $v = B_q a + qzv$, $b = D_q a + B_q^T zv$. Elimination gives

$$V_q(z) = \frac{1}{1-qz} \begin{pmatrix} 1-q & \sqrt{q}(1-z) \\ \sqrt{q}(1-z) & (1-q)z \end{pmatrix}, \quad \det V_q(z) = \frac{z-q}{1-qz}. \quad (12.1)$$

The matrix is unitary for $|z| = 1$. Its complementary outputs for one incident lead are $K_0 = (1-q)/(1-qz)$ and $K_1 = \sqrt{q}(1-z)/(1-qz)$, with $|K_0|^2 + |K_1|^2 = 1$.

12.2 The prime source and its compulsory contact

Feed the Fourier input into the complementary port with prescribed scale $\sqrt{d(1+q)/(1-q)}$, and attach the unit cap χ if desired. Equivalently, with $U = U_d$, the bounded whole-line source is

$$\mathcal{D}_p F = \sqrt{\frac{dq(1+q)}{1-q}} (I - U)(I - qU)^{-1} F. \quad (12.2)$$

The inverse is its norm-convergent Neumann series. Two graph copies fed by $\hat{F}(k)$ and $\hat{F}(-k)$ accommodate the two Fourier signs with nonnegative graph energy k^2 . The arithmetic Fourier variable is a wave number in this construction, not a physical evolution time. The Hilbert adjoint of (12.2) follows by replacing U by U^* and reversing the factors, and the interval adjoint includes E_L^* .

The Poisson kernel

$$\Pi_q(z) = \frac{1 - q^2}{|1 - qz|^2} = 1 + \sum_{m \geq 1} q^m (z^m + z^{-m})$$

gives the exact output identity

$$\mathcal{D}_p^* \mathcal{D}_p = \frac{d(1+q)}{1-q} I - d \Pi_q(U) = c_p I - d \sum_{m \geq 1} q^m (U^m + U^{-m}), \quad c_p = \frac{2dq}{1-q}. \quad (12.3)$$

Moreover $\|\mathcal{D}_p\|^2 = 4dq/(1-q^2)$. Compression deletes precisely those terms with $md \geq L$. Thus the channel gives every repetition with the arithmetic weight, but also a positive contact c_p . The incident scale was prescribed to obtain those weights; vertex unitarity alone does not choose that scale.

Proposition 12.1. *For a finite prime set $\mathcal{S}$, a stationary whole-line Gram operator with off-diagonal part $-\sum_{p \in \mathcal{S}} d_p \sum_{m \geq 1} q_p^m (U_{md_p} + U_{-md_p})$ and constant diagonal cI is nonnegative if and only if $c \geq c_{\mathcal{S}} := \sum_{p \in \mathcal{S}} c_p$.*

Proof. Each cosine in the multiplier is at most one, which proves sufficiency. At frequency zero they are simultaneously one. Fourier packets concentrated at zero prove necessity. The argument applies to arbitrary coherent realizations of the same stationary Gram operator. $\square$

This lower bound is for the whole-line stationary model. At fixed support a smaller contact can suffice. For one prime, decompose I_L into translation chains modulo d . The largest chain has $n = \lceil L/d \rceil$ points, almost everywhere when this length occurs, and the correlation matrix on such a chain is $R_n(q) = (q^{|i-j|})_{i,j=1}^n$. The exact interval contact is

$$c_{p,L}^{\min} = d(\lambda_{\max}(R_n(q)) - 1). \quad (12.4)$$

It is zero for $L \leq d$ and dq for $d < L \leq 2d$. Testing the long chain on the constant vector, together with the row-sum upper bound, shows that it tends to c_p as $L \rightarrow \infty$.

12.3 Phase derivative and the Euler logarithm

For $Z(s) = (1 - p^{-s})^{-1}$ on $\text{Re } s = 1/2$,

$$2 \text{Re } \partial_s \log Z(s) = d - d \Pi_q(e^{-i d \text{Im } s}).$$

Its contact is therefore different from the output-power contact in (12.3). In contrast, the scattering phase satisfies

$$\partial_k \arg \det V_q(e^{ikd}) = d \Pi_q(e^{ikd}) > 0, \quad -i V_q^\dagger \partial_k V_q = \frac{d}{|1 - q e^{ikd}|^2} B_q^T B_q \geq 0.$$

The trace is the phase derivative, and differentiation with respect to energy k^2 introduces $1/(2k)$ for $k > 0$. These formulas distinguish Euler logarithmic data, positive output power and positive delay; none identifies them with the complete arithmetic norm.

For a fixed interval it is sufficient to take $\mathcal{S}_L = \{p : \log p < L\}$. An unrenormalized direct sum over all primes fails even on a nonzero compactly supported input. For the inactive primes all translates are disjoint from the input, so their output norm is exactly $c_p \|f\|^2$; the sum of c_p diverges (already it dominates a constant multiple of $\sum_p 1/p$). The finite-prime graph therefore does not itself define an all-prime positive limit.

13 Collective feedback and its compact error

The delay source A_0 in (11.3) is closed on $\mathcal{D}_{\log,L}$ and has $T_0 = A_0^* A_0 \geq mI$. It permits a stable response to a prescribed input. This is the first step toward replacing compulsory contact by a controlled error.

13.1 One response field

For $\mu \geq 0$, minimize

$$\|A_0 u\|^2 + \mu \|u\|^2 - 2 \operatorname{Re} \langle f, u \rangle \quad \text{over } u \in \mathcal{D}_{\log,L}.$$

This is stable for every fixed f and has minimizer $u = R_\mu f$, $R_\mu = (T_0 + \mu)^{-1}$. Stability here does not assert joint positivity in (f, u) . Read out $A_{\text{fb}} = A_0(I - \kappa R_\mu)$, $\kappa > 0$. The correction $F_{\kappa,\mu} = -\kappa A_0 R_\mu$ is bounded and compact, so the source remains closed with the same domain and maximal adjoint domain. Expanding all terms, including the feedback self-pairing, yields

$$A_{\text{fb}}^* A_{\text{fb}} = T_0 - 2\kappa I + C_{\kappa,\mu}, \quad C_{\kappa,\mu} = 2\kappa\mu R_\mu + \kappa^2 T_0 R_\mu^2. \quad (13.1)$$

This error is positive, compact and infinite rank.

Let $J = \bigoplus_{p \in \mathcal{S}_L} \mathcal{D}_p E_L$ and $c_{\mathcal{S}} = \sum_{p \in \mathcal{S}_L} c_p$. Choose the gain by

$$2\kappa = e^D + c_{\mathcal{S}} - w_0. \quad (13.2)$$

Then the assembled source $\Gamma_{\kappa,\mu} = (A_{\text{fb}}, J)$ satisfies

$$\|\Gamma_{\kappa,\mu} f\|^2 = Q_L[f] + \langle f, C_{\kappa,\mu} f \rangle. \quad (13.3)$$

Its block charge is a closed-source supersymmetric construction, and its form operator is $W_L + C_{\kappa,\mu}$ on $\operatorname{Dom} T_L$. The gain and prime scales match arithmetic coefficients; the positive response has not independently derived those parameters. The quartic cap is an isometric tensor factor in this identity.

13.2 The error cannot be removed by its regulator

On a T_0 eigenvalue λ , the error equals

$$c_{\kappa,\mu}(\lambda) = \frac{2\kappa\mu}{\lambda + \mu} + \frac{\kappa^2\lambda}{(\lambda + \mu)^2}, \quad \partial_\mu c_{\kappa,\mu}(\lambda) = \frac{2\kappa\lambda(\lambda + \mu - \kappa)}{(\lambda + \mu)^3}.$$

In particular,

$$\inf_{\mu \geq 0} c_{\kappa,\mu}(\lambda) = \begin{cases} \kappa^2/\lambda, & \lambda \geq \kappa, \\ 2\kappa - \lambda, & 0 < \lambda < \kappa. \end{cases}$$

Every fixed mode retains positive error. As $\mu \downarrow 0$, the operator converges in norm to $\kappa^2 T_0^{-1}$; as $\mu \rightarrow \infty$, it converges strongly to $2\kappa I$, but not in norm because it is compact for every finite μ . For fixed parameters, $c_{\kappa,\mu}(\lambda) \sim (\kappa^2 + 2\kappa\mu)/\lambda$. Compactness alone gives no smallness on the possibly negative arithmetic modes.

13.3 Schatten-class obstruction

Choose a subinterval J_0 of length $\ell < \min(L, \log 2)$ and the orthonormal inputs $f_n = \ell^{-1/2} 1_{J_0} e^{2\pi i n x / \ell}$. They belong to $\mathcal{D}_{\log, L}$ by their sinc Fourier tails. All prime overlaps vanish, the pole contribution is $O(n^{-2})$, and $\|A_L f_n\|^2 = O(\log n)$. Therefore $Q_L[f_n] - \|A_L f_n\|^2 = w_0 + O(n^{-2})$. If a correction B made $\|(A_L + B)f\|^2 = Q_L[f]$ on the full domain, then Cauchy–Schwarz in its cross term would force

$$\|B f_n\|^2 \geq c / \log n \quad \text{for all sufficiently large } n.$$

This is incompatible with B in any finite Schatten class: for $p \geq 2$, convexity and the trace bound give $\sum_n \|B f_n\|^p \leq \sum_n s_n(B)^p$; for $p < 2$ use inclusion in the Hilbert–Schmidt class. The same reasoning applies to the assembled reference (A_0, J) , whose target discrepancy is exactly the nonzero constant $-2\kappa I$. It excludes finite Schatten-class source corrections, not every compact correction.

The spectral comparison sharpens this distinction. Since P_L has rank two, finite-rank interlacing and (10.11) give

$$\lambda_n(T_0) = \log \frac{\pi n}{2L} - \psi(1/4) + e^D + O_L(n^{-1}). \quad (13.4)$$

Consequently the decreasing singular values satisfy

$$\lim_{n \rightarrow \infty} \sqrt{\log n} s_n(F_{\kappa, \mu}) = \kappa, \quad \lim_{n \rightarrow \infty} \log n s_n(C_{\kappa, \mu}) = \kappa^2 + 2\kappa\mu. \quad (13.5)$$

Neither lies in any finite Schatten class. These exact leading scales replace the earlier one-sided logarithmic comparison.

14 Joint response and an exact finite-dimensional remainder

The compact error in Section 13 can be replaced by a finite-rank error. The construction uses a proved positive high sector and keeps every coupling to its finite-dimensional complement. It leads to an exact finite matrix criterion, whose sign remains to be established.

14.1 Choose the high sector without the Weil spectrum

For $q \in (0, 1)$ put

$$r_n(q) = \max_{1 \leq i \leq n} \left(\sum_{j=1}^{i-1} q^j + \sum_{j=1}^{n-i} q^j \right), \quad n_p = \lceil L / \log p \rceil.$$

The row-sum bound on the finite prime chains in (12.4), the negative odd pole bound and Theorem 10.2 give

$$W_L \geq B(H_N) - \beta_L I, \quad \beta_L = -w_0 + 2 \sinh(L/2) - L + \sum_{\log p < L} (\log p) r_{n_p}(p^{-1/2}). \quad (14.1)$$

The coarser replacement of the last sum by $2 \sum_{m \log p < L} (\log p) p^{-m/2}$ is also valid.

Use the explicit Neumann modes

$$e_0 = L^{-1/2}, \quad e_j(x) = \sqrt{2/L} \cos(j\pi(x + L/2)/L) \quad (j \geq 1), \quad P = P_N = \sum_{j=0}^{N-1} e_j e_j^*, \quad Q = I - P.$$

Choose $N \geq 1, \delta > 0$ such that

$$B((N\pi/L)^2) \geq \beta_L + \delta. \quad (14.2)$$

Such choices exist at each fixed L . The restricted closed arithmetic form then satisfies $W_L|_{QX_L} \geq \delta I$. This is not a statement about the mixed low/high terms. The vectors e_j belong even to $\text{Dom } T_L$: their zero-extension Fourier transforms are $O_j((1+|\tau|)^{-1})$, which remain square-integrable after multiplication by $B(\tau^2)$. Thus P preserves the form domain and QW_LP is bounded and finite rank.

There is a wholly rational example at $L = 1$. Only prime two's first return is active, and

$$\beta_1 < \frac{58}{100} + \frac{11}{7} + \frac{21}{10} + \frac{23}{20} + \frac{43}{1000} + \frac{1}{2} = \frac{41611}{7000}.$$

The bounds used here are $\gamma_E < 0.58$, $\pi < 22/7$, $\log 2 < 0.7$, $\log \pi < 23/20$, $2 \sinh(1/2) - 1 < 0.043$, and $\log 2/\sqrt{2} < 1/2$. They follow from positive or alternating rational series. For example $\log 2 = 2 \sum_{j \geq 0} 3^{-(2j+1)}/(2j+1) > 6931/10000$, so $\gamma_E < H_{256} - 8(6931/10000) < 58/100$. The integral identities for $\arctan 1$ and $22/7 - \pi = \int_0^1 x^4(1-x)^4/(1+x^2) dx$ give $3 < \pi < 22/7$. A finite exponential sum bounds $e^{23/20}$ from below by $22/7$; a geometric tail bounds the positive sinh series. Finally, exact fraction arithmetic gives

$$B((4\pi)^2) > \sum_{k=0}^{63} \frac{2}{a_k} \frac{144}{a_k^2 + 144} > \frac{3007}{500} > \frac{41611}{7000} + \frac{1}{16}.$$

Thus $N = 4$, $\delta = 1/16$ is a proved high-mode cutoff. It is not a certificate for the full form at $L = 1$.

14.2 One field drives all channels

Combine the already specified sources in the same positive space:

$$A = (A_0, J), \quad T = A^*A = T_0 + J^*J = W_L + \alpha I, \quad \alpha = e^D + c_S - w_0 = 2\kappa, \quad T \geq mI. \quad (14.3)$$

The domain is $\mathcal{D}_{\log, L}$ and $\text{Dom } T = \text{Dom } T_L$. Prepare a common field in both the gamma-and-pole source and every prime graph by minimizing $\|A_0 u\|^2 + \|Ju\|^2 - 2 \text{Re} \langle v, u \rangle$. Its response is $T^{-1}v$. A first joint readout obeys

$$\|A(I - \alpha T^{-1}/2)f\|^2 = Q_L[f] + \frac{\alpha^2}{4} \langle f, T^{-1}f \rangle.$$

Since $T \geq T_0 > 0$, this error is no greater than the earlier massless error. The full repeated response gives a stronger result.

Restrict A to $Q\mathcal{D}_{\log, L}$, denoting it A_H , and put

$$T_H = A_H^* A_H, \quad V = T_H^{-1}, \quad H = T_H - \alpha I \geq \delta I, \quad B = QTP = QW_LP, \quad M = PTP.$$

In this section B without an argument is the finite coupling matrix; $B(s)$ remains the gamma multiplier. The arithmetic content of the coupling is explicit:

$$B = QT_LP + QP_LP - \sum_{m \log p < L} (\log p) p^{-m/2} Q(T_{m \log p} + T_{m \log p}^*)P.$$

For $f = p + q$, with $p = Pf$, $q = Qf$, the exact form is

$$Q_L[p + q] = \langle q, Hq \rangle + 2 \text{Re} \langle q, Bp \rangle + \langle p, (M - \alpha I)p \rangle. \quad (14.4)$$

Terms involving H are interpreted as forms on its form domain.

14.3 A convergent response series with all mixed returns

Let

$$a_j = \frac{\binom{2j}{j}}{4^j(2j-1)}, \quad a_1 = \frac{1}{2}, \quad a_{j+1} = a_j \frac{2j-1}{2j+2}, \quad \theta = \frac{\alpha}{\alpha + \delta} < 1.$$

The operator series

$$R = I - \sum_{j \geq 1} a_j \alpha^j V^j, \quad R_\ell = I - \sum_{j=1}^{\ell} a_j \alpha^j V^j \quad (14.5)$$

converges in norm and gives a positive invertible R with

$$R^2 = I - \alpha V, \quad RT_H R = H, \quad \|A_H R q\|^2 = \langle q, H q \rangle.$$

It is precisely the positive high-sector factor $(I - \alpha V)^{1/2}$. Its use is justified by the independent bound (14.2); no sign of the full W_L has been assumed. The stronger estimate needed for the unbounded output is

$$\|A_H(R_\ell - R)\| \leq \frac{a_{\ell+1} \alpha^{\ell+1}}{(1 - \theta)(\alpha + \delta)^{\ell+1/2}} \rightarrow 0. \quad (14.6)$$

Indeed the tail on an eigenvalue $\lambda \geq \alpha + \delta$ is $\sqrt{\lambda} \sum_{j > \ell} a_j (\alpha/\lambda)^j$. Moreover

$$E_\ell := T_H R_\ell^2 - H \geq 0, \quad \|E_\ell\| \leq \frac{2a_{\ell+1} \alpha \theta^\ell}{1 - \theta}, \quad \ell \geq 1. \quad (14.7)$$

Each E_ℓ is compact. A small ratio δ/α can make this convergence slow.

To specify the noncommuting returns, write $T_H = T_{0,H} + K_H$ with $K_H = (J|_Q)^*(J|_Q) \geq 0$. Choose the known bound $t = \sum_{p \in \mathcal{S}} 4d_p q_p / (1 - q_p^2) \geq \|K_H\|$ and put $V_0 = (T_{0,H} + tI)^{-1}$. Then

$$V = \sum_{j \geq 0} [V_0(tI - K_H)]^j V_0, \quad \|V_0(tI - K_H)\| \leq \frac{t}{m + t} < 1, \quad (14.8)$$

with just V_0 when $t = 0$. Here

$$K_H = c_S I_{QX_L} - \sum_{m \log p < L} (\log p) p^{-m/2} Q (T_{m \log p} + T_{m \log p}^*) Q.$$

The order of the intervening resolvents and compressions in (14.8) retains every mixed gamma/prime return. Replacing these ordered words by scalar Euler multipliers would change the operator. Also $H^{-1} = \sum_{j \geq 0} \alpha^j V^{j+1}$ converges in norm, supplying the second high-sector response below.

14.4 Retain the unused output and obtain finite-rank error

The range of A_H is closed. Its orthogonal projection is the bounded extension $\Pi_H = A_H V A_H^*$. On a low input,

$$Ap = A_H V B p + Z p, \quad Z = (I - \Pi_H) A|_{PX_L}, \quad G := Z^* Z = M - B^* V B \geq mI.$$

The last bound follows by minimizing the known positive T -energy over the high input at fixed p . In particular no unknown arithmetic Schur complement has been declared positive.

Theorem 14.1. *In the original positive output space, define*

$$\Phi_L(p + q) = A_H R(q + H^{-1} Bp) + Zp. \quad (14.9)$$

It is a closed source on $\mathcal{D}_{\log, L}$, and for all inputs in that domain its polarized Gram identity is

$$\langle \Phi_L f, \Phi_L g \rangle = Q_L(f, g) + \langle Pf, D_N P g \rangle, \quad D_N = \alpha I + B^*(H^{-1} - V)B = \alpha(I + B^* V H^{-1} B) > 0. \quad (14.10)$$

The error operator $PD_N P$ has rank exactly N .

Proof. The two outputs in (14.9) are orthogonal. Expanding their norm with $RT_H R = H$ gives $\langle q, Hq \rangle + 2 \operatorname{Re} \langle q, Bp \rangle + \langle p, (G + B^* H^{-1} B)p \rangle$. Subtract (14.4), use $G = M - B^* V B$, and use $H^{-1} - V = \alpha V H^{-1} \geq 0$ to obtain (14.10). Polarization gives the bilinear identity. For closedness subtract the original source:

$$\Phi_L - A = A_H(R - I)Q + A_H(RH^{-1} - V)BP =: F. \quad (14.11)$$

The first term extends to a compact bounded operator since its singular magnitude at $\lambda \in \operatorname{spec} T_H$ is

$$\sqrt{\lambda}(1 - \sqrt{1 - \alpha/\lambda}) = \frac{\alpha}{\sqrt{\lambda}(1 + \sqrt{1 - \alpha/\lambda})} \rightarrow 0.$$

The second is bounded and finite rank. Thus $\Phi_L = A + F$ is a bounded perturbation of a closed source. $\square$

It has the same graph topology and smooth compact-support core as A , and its maximal adjoint is $\Phi_L^* = A^* + F^*$ on $\operatorname{Dom} A^*$. Consequently $\Phi_L^* \Phi_L = W_L + PD_N P$ on $\operatorname{Dom} T_L$. Using R_ℓ instead of R gives the exact additional error

$$(Q + H^{-1} BP)^* E_\ell (Q + H^{-1} BP) \geq 0,$$

which tends to zero in operator norm by (14.7). Finite series generally leave an infinite-rank extra error.

The correction F itself is not finite rank. Bounded-perturbation min-max and finite-codimension interlacing give $\lambda_n(T_H) = \log n + O_L(1)$. The displayed singular magnitude and finite-rank interlacing in (14.11) imply

$$\sqrt{\log n} s_n(F) \rightarrow \alpha/2. \quad (14.12)$$

Thus the source correction remains outside every finite Schatten class, consistently with Section 13. Finite rank of the form error does not imply finite rank of the source change.

14.5 The exact remaining matrix criterion

Completing the high square in (14.4) gives

$$Q_L[p + q] = \|A_H R(q + H^{-1} Bp)\|^2 + \langle p, S_N p \rangle, \quad S_N = M - \alpha I - B^* H^{-1} B = G - D_N. \quad (14.13)$$

Since $q = -H^{-1} Bp$ is admissible for every p ,

$$Q_L \geq 0 \iff S_N \geq 0 \iff G^{-1/2} D_N G^{-1/2} \leq I. \quad (14.14)$$

Both G and D_N are known positive matrices; their difference has no established sign. At $L = 1$ the rational cutoff above makes this a four-by-four comparison, but no entries or positivity certificate for that arithmetic matrix are asserted.

A rational non-arithmetic control makes the logical limit explicit. Take $\alpha = 1$,

$$T_H = \text{diag}(25/16, 25/9), \quad H = \text{diag}(9/16, 16/9), \quad B = \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}, \quad Z_c = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}.$$

On two low and two high coordinates use the invertible source

$$A = \begin{pmatrix} T_H^{-1/2} B & T_H^{1/2} \\ Z_c & 0 \end{pmatrix}, \quad \Phi = \begin{pmatrix} H^{-1/2} B & H^{1/2} \\ Z_c & 0 \end{pmatrix}.$$

Their Gram identity holds exactly, with

$$D_N = \frac{1}{3600} \begin{pmatrix} 14257 & 10379 \\ 10379 & 20713 \end{pmatrix}, \quad S_N = \frac{1}{3600} \begin{pmatrix} 143 & -3179 \\ -3179 & -13513 \end{pmatrix}.$$

The vector $p = (0, 1)$, $q = (-32/9, -9/16)$ has $\langle p + q, (A^* A - I)(p + q) \rangle = -13513/3600 < 0$. Thus a positive shifted source, a positive high sector and the completion identity can coexist with a negative full target. The reduction is valid for a wider class of semibounded forms; its existence alone cannot be an arithmetic positivity proof.

14.6 Compatibility inside a fixed support envelope

Construct Φ_* once on I_{L*} , fixing its delay, prime set and mode projection. For $L \leq L_*$ let j_L be zero extension into that envelope. The restricted sources are closed on the full logarithmic domains, have the same smooth cores, and obey

$$\|\Phi_* j_L f\|^2 = Q_L[f] + \langle f, j_L^* P_* D_* P_* j_L f \rangle. \quad (14.15)$$

The complete arithmetic form is unchanged under the extension; the additional inactive prime overlaps vanish. These sources are compatible with nested intervals inside the fixed envelope. Rebuilding the source on a larger envelope generally changes the compressed error. No single all-support source or all-prime limit, and no rule allowing these errors to be subtracted, has been obtained.

15 What the remaining physical-pairing problem now requires

The investigation now has two substantial constructions. The finite-dimensional one gives genuine physical caps, their residue normalization, a normalized interacting Euler source and controlled endpoint geometry. The infinite-dimensional one gives a closed preparation for the complete arithmetic form plus an explicit finite-rank positive error. The bridge between them is still incomplete: the unit quartic cap can label the response channels isometrically, but the resulting norm identities remain the same if that label is suppressed.

The exact remaining arithmetic question at fixed support is

$$G - D_N \geq 0$$

in the finite cosine sector of Section 14. Both matrices arise from explicit sources and stable responses. Their positivity separately is proved; their ordering is not. Further iteration of the already solved high-sector series cannot determine this ordering. A useful continuation would bound the mixed response $B^* H^{-1} B$ using the arithmetic structure of the ordered return expansion, or

derive a physical constraint that identifies the finite difference as a positive Gram pairing without assuming its sign.

A verified enclosure of the matrix at one length would be a finite-support result. The larger problem requires control under support enlargement. The compatible restriction law (14.15) works inside one fixed envelope; rebuilding on a larger envelope changes the finite error. No all-support limit or cancellation rule for those errors has been proved. The divergence of unrenormalized all-prime output power remains an additional constraint on such a law.

On the physical metric side, the outstanding identification is the exact chiral equation for the harmonic family in the chosen source frame. Once that identification is established, the endpoint theorem gives uniqueness and implicitly selects the remaining homogeneous scale. Computing that scale or the radial metric could illuminate the geometry, but would not by itself settle the arithmetic finite-matrix sign. The periodic gamma sector also still requires the global deck or flavor data for its multivalued insertion and a calculation of the relevant physical arithmetic pairing.

The scoped obstructions remain useful tests. Scalar transforms, finite ambient-holomorphic vacuum enlargement, bounded or closable finite-range source maps, the specified finite-delay class and finite Schatten-class source corrections each fail under their stated assumptions. The joint source avoids the last obstruction by a compact infinite-rank correction with singular values asymptotic to $\alpha/(2\sqrt{\log n})$. Its finite-rank form error must still be retained. The complete positive Weil norm, an all-support positivity theorem and a proof of the Riemann hypothesis remain open.

15.1 Reproducibility and review scope

Nine small standard-library programs now replay 49 labelled exact algebra checks. They cover the common-core supercharge relations, quartic and shape period elimination, residue and reflection normalization, endpoint matrices, Robin and delay identities, prime scattering, collective feedback, noncommuting joint-response blocks and the rational high-mode cutoff. The programs do not certify analytic domains, Hodge theorems, asymptotic limits, or the sign of the arithmetic Schur matrix. Those arguments and hypotheses are stated separately here.

The source package contains a coverage map from each investigation note to this integrated manuscript, the TeX build record, the exact numerical records and a replay validator. The preceding 25-page draft is preserved byte for byte with its TeX sources and build record under `archive/drafts/20260913-pre-integration/`. The integrated version is also saved with its complete sources under `archive/drafts/20260913-integrated/`. The earlier draft's historical claims and records have not been rewritten to make them appear to contain the later results. The package manifest records file identities; it is not a mathematical proof certificate.

This is an AI-assisted working manuscript. Internal analytical review and exact algebra replay do not replace independent specialist review or establish a priority claim.

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